

GHA & Declination to almanac accuracy on the C47

Two methods, C47 (HP-42S-style) programs and data for 1 Sep 2026 – 31 Dec 2027

Summary

Both methods reproduce the Nautical Almanac to better than its 0.1' printing resolution. All values are **apparent, geocentric** positions, referred to the true equator and equinox of date. The time argument is **UT (UT1)**, as in the almanac.

	Method A — series (Sun only)	Method B — Chebyshev tables
What the C47 does	Computes everything from time: truncated VSOP87 Earth theory + 10-term IAU 1980 nutation + sidereal time	Evaluates a short polynomial whose coefficients you key in from the tables in this document
Bodies	Sun and Aries	Sun, Aries, Moon (incl. HP), Venus, Mars, Jupiter, Saturn
Validity	Any date (tested 2025–2035); degrades slowly beyond ± 100 years	1 Sep 2026 – 31 Dec 2027 (regenerate for later years)
Data to enter once	66 VSOP rows (4 numbers each) + 10 nutation rows (9 numbers)	Nothing permanent
Data per sight	None	10–16 numbers (one table line)
Max. error vs reference	GHA 0.011', Dec 0.006' (Aries 0.002')	GHA and Dec $\leq 0.002'$; Moon HP $\leq 0.001'$

How the reference was built and checked

The coefficient tables were fitted to a full-precision reference: JPL DE421 ephemeris; IAU 2006/2000A precession-nutation and Earth Rotation Angle (SOFA/ERFA); light-time, annual aberration and solar light deflection; $\Delta T = TT - UT1 = 69.2$ s. The same models are used by USNO and HMNAO. Checked against the USNO Celestial Navigation Data service:

Comparison with USNO service	Difference
Sun, 23 Sep 2026 12h and 10 Mar 2027 05:30	GHA $\leq 0.007'$, Dec $\leq 0.003'$
Aries (GHA Υ), same dates	0.000'
Mars, Jupiter, Saturn	$\leq 0.004'$
Moon, 23 Sep 2026 19h	GHA 0.085', Dec 0.04'. All of this is from ΔT : the USNO web service appears to use $\Delta T \approx 80$ s; with 80 s the difference drops to 0.004'. The Moon moves about 0.55' per second of ΔT . These tables use the measured value, 69.2 s.
Venus near inferior conjunction (Sep 2026)	About 0.22'. The USNO value is offset towards the Sun, which looks like a phase (centre-of-light) correction. These tables give the centre of the disk. Far from conjunction (Mar 2027) the difference is 0.04'.

Using UTC instead of UT1: the tables and programs take UT1. For a sight timed in UTC, add DUT1 (broadcast with time signals, $|DUT1| < 0.9$ s). Ignoring it costs up to 0.23' in GHA, the same as when using the printed almanac.

About the program listings

The listings use HP-42S mnemonics. The C47 keeps the 42S programming model, but some function names on its keyboard and menus may differ. Check these on your calculator before keying in: **RCL+** **RCL–** **RCL $\times$** **RCL $\div$** (recall arithmetic), **RCL IND** and **DSE** (indirect registers and loop counter ccc.fffii), **→POL** (rectangular to polar), **MOD**, and the matrix commands **INDEX**, **STOIJ**, **RCLEL**, **J+**. SUNA has been run on the C47 and reproduces the check values below. It converts radians with $\times 57.29577951\dots$ rather than $\rightarrow\text{DEG}$, and its loops count rows with DSE 55 rather than testing flag 77, because those two behave differently on the C47. Every listing was run line by line in a 34-digit RPN simulator and matched the Python model to better than 0.00001'.

Method A — Sun from truncated VSOP87

This is Meeus, Astronomical Algorithms, Ch. 22, 25 and 32, with the VSOP87D Earth series truncated to the 66 terms that matter at the 0.01' level. Enter the Julian Date (UT1) in X and run **SUNA**. It returns X = GHA ☉, Y = Dec ☉, Z = GHA ♀.

Formulas (what the program does)

#	Step	Formula
1	Time	$JDE = JD + 69.2/86400; T = (JDE - 2451545)/36525; \tau = T/10$
2	Earth heliocentric	$L = \sum \tau^k \cdot A \cdot \cos(B + C \cdot \tau) \times 10^{-8}$ (rad), same for B and R (matrices VL, VB, VR)
3	Geocentric Sun	$\odot = L + 180^\circ, \beta = -B$
4	FK5 correction	$L' = \odot - 1.397^\circ \cdot T; \odot - = 0.09033''; \beta + = 0.03916'' \cdot (\cos L' - \sin L')$
5	Nutation	D, M, M', F, Ω linear in T (Meeus 22); $\arg = \sum m_i \cdot \arg_i; \Delta\psi = \Sigma(S + S'T) \cdot \sin \arg; \Delta\epsilon = \Sigma(C + C'T) \cdot \cos \arg$ (0.0001")
6	Apparent longitude	$\lambda = \odot + \Delta\psi - 20.4898''/R$
7	Obliquity	$\epsilon = 84381.448'' - 46.815'' \cdot T + \Delta\epsilon$
8	Equatorial	$\text{Dec} = \text{asin}(\sin \beta \cos \epsilon + \cos \beta \sin \epsilon \sin \lambda); \text{RA} = \text{atan2}(\sin \lambda \cos \epsilon - \tan \beta \sin \epsilon, \cos \lambda)$
9	Sidereal time	$d = JD(UT1) - 2451545; \text{GMST} = 280.46061837 + 360.98564736629 \cdot d + 0.000387933 \cdot (d/36525)^2; \text{GHA } \Upsilon = \text{GMST} + \Delta\psi \cdot \cos \epsilon$
10	GHA	$\text{GHA } \odot = \text{GHA } \Upsilon - \text{RA} \pmod{360}$

Check values — 23 Sep 2026, 12:00:00 UT1 (JD 2461307.0)

Quantity	Value	Quantity	Value
T	0.26726901577	Δψ	+8.8954"
τ	0.026726901577	Δε	+8.3864"
L (rad, unreduced)	169.654534791	ε	23.4381451°
B (rad)	+2.524e-7	λ (apparent)	180.4855926°
R (au)	1.00344507	RA	180.4455179°
GHA ♀	182.3525029° = 182° 21.15'	Dec ☉	-0.1931700° = S 0° 11.59'
GHA ☉	1.9069849° = 1° 54.42' USNO (for comparison)		1° 54.41' / S 0° 11.59'

Registers used

R50 τ · R51–R53 series work · R54 T · R60–R67 nutation · R70 JD · R71 L · R72 B · R73 R · R74 λ · R75 β · R76 ε · R77 Dec · R78 RA · R79 d · R80 GHA ♀ · R81 GHA ☉. Matrices: **VL** 56×4, **VB** 3×4, **VR** 7×4, **NU** 10×9 (create with e.g. 56 ENTER 4 NEWMAT, STO "VL", then fill with EDITN).

Program SUNA (with subroutines SER and NUT)

001	LBL "SUNA"	; X=JD(UT1)	073	STO- 74	; lambda	145	RCL 77	
002	DEG		074	84381.448		146	RCL 81	; X,Y,Z
003	STO 70		075	RCL 54		147	RTN	
004	69.2		076	46.815		148	LBL "SER"	; series
005	86400		077	x		149	0	
006	÷		078	-		150	STO 51	
007	+		079	RCL+ 67		151	1	
008	2451545		080	3600		152	1	
009	-		081	÷		153	STOIJ	
010	36525		082	STO 76	; eps	154	LBL 12	
011	÷		083	RCL 75	; Dec	155	RCLEL	
012	STO 54	; T	084	SIN		156	J+	
013	10		085	RCL 76		157	STO 52	
014	÷		086	COS		158	RCLEL	
015	STO 50	; tau	087	x		159	J+	
016	RAD		088	RCL 75		160	STO 53	
017	56		089	COS		161	RCLEL	
018	STO 55		090	RCL 76		162	J+	
019	INDEX "VL"		091	SIN		163	RCLEL	
020	XEQ "SER"		092	x		164	J+	
021	STO 71	; L (rad)	093	RCL 74		165	RCLx 50	
022	3		094	SIN		166	+	
023	STO 55		095	x		167	COS	
024	INDEX "VB"		096	+		168	RCLx 53	
025	XEQ "SER"		097	ASIN		169	RCL 50	
026	STO 72	; B (rad)	098	STO 77		170	RCL 52	
027	7		099	RCL 74	; RA	171	Y^X	
028	STO 55		100	SIN		172	x	
029	INDEX "VR"		101	RCL 76		173	STO+ 51	
030	XEQ "SER"		102	COS		174	DSE 55	; next row
031	STO 73	; R (au)	103	x		175	GT0 12	
032	DEG		104	RCL 75		176	RCL 51	
033	RCL 71		105	TAN		177	1E-8	
034	57.2957795130823208767981548141006		106	RCL 76		178	x	
035	x	; rad→deg	107	SIN		179	RTN	
036	180		108	x		180	LBL "NUT"	; nutation
037	+		109	-		181	RCL 54	
038	360		110	RCL 74		182	445267.11148	
039	MOD		111	COS		183	x	
040	STO 74	; lon	112	→POL		184	297.85036	
041	RCL 72		113	X<=>Y		185	+	
042	57.2957795130823208767981548141054		114	STO 78		186	STO 60	
043	x	; rad→deg	115	RCL 70	; GAST	187	RCL 54	
044	CHS		116	2451545		188	35999.05034	
045	STO 75	; lat	117	-		189	x	
046	RCL 74	; FK5	118	STO 79		190	357.52772	
047	RCL 54		119	360.98564736629		191	+	
048	1.397		120	x		192	STO 61	
049	x		121	280.46061837		193	RCL 54	
050	-		122	+		194	477198.867398	
051	COS		123	RCL 79		195	x	
052	LASTX		124	36525		196	134.96298	
053	SIN		125	÷		197	+	
054	-		126	X^2		198	STO 62	
055	0.03916		127	0.000387933		199	RCL 54	
056	x		128	x		200	483202.017538	
057	3600		129	+		201	x	
058	÷		130	RCL 66		202	93.27191	
059	STO+ 75		131	3600		203	+	
060	0.09033		132	÷		204	STO 63	
061	3600		133	RCL 76		205	RCL 54	
062	÷		134	COS		206	1934.136261	
063	STO- 74		135	x		207	x	
064	XEQ "NUT"	; R66,R67	136	+		208	CHS	
065	RCL 66		137	360		209	125.04452	
066	3600		138	MOD		210	+	
067	÷		139	STO 80	; GHA Ari	211	STO 64	
068	STO+ 74		140	RCL- 78		212	0	
069	20.4898		141	360		213	STO 66	
070	RCL÷ 73		142	MOD		214	STO 67	
071	3600		143	STO 81	; GHA Sun	215	10	
072	÷		144	RCL 80		216	STO 55	

217	INDEX "NU"	236	+	255	J+
218	1	237	RCLEL	256	RCL× 54
219	1	238	J+	257	+
220	STOIJ	239	RCL× 64	258	RCL 65
221	LBL 13	240	+	259	COS
222	RCLEL	241	STO 65 ; arg	260	×
223	J+	242	RCLEL	261	STO+ 67
224	RCL× 60	243	J+	262	DSE 55 ; next row
225	RCLEL	244	RCLEL	263	GT0 13
226	J+	245	J+	264	RCL 66
227	RCL× 61	246	RCL× 54	265	1E-4
228	+	247	+	266	×
229	RCLEL	248	RCL 65	267	STO 66
230	J+	249	SIN	268	RCL 67
231	RCL× 62	250	×	269	1E-4
232	+	251	STO+ 66	270	×
233	RCLEL	252	RCLEL	271	STO 67
234	J+	253	J+	272	RTN
235	RCL× 63	254	RCLEL		

Matrix VL — Earth heliocentric longitude L (56 rows)

Columns: k (power of τ), A ($\times 10^{-8}$ rad), B (rad), C (rad per millennium). Values from VSOP87D (Bretagnon & Francou 1988).

Row	k	A	B	C	Row	k	A	B	C
1	0	175347045.673	0.00000000000	0.00000000	29	0	115.132	0.64544911683	0.98032107
2	0	3341656.456	4.66925680417	6283.07584999	30	0	102.851	0.63599846727	4694.00295471
3	0	34894.275	4.62610241759	12566.15169998	31	0	101.724	4.26679821365	7.11354700
4	0	3417.571	2.82886579606	3.52311835	32	0	99.206	6.20992940258	2146.16541648
5	0	3497.056	2.74411800971	5753.38488490	33	0	132.212	3.41118275555	2942.46342329
6	0	3135.896	3.62767041758	77713.77146812	34	0	97.607	0.68101272270	155.42039943
7	0	2676.218	4.41808351397	7860.41939244	35	0	85.128	1.29870743025	6275.96230299
8	0	2342.687	6.13516237631	3930.20969622	36	0	74.651	1.75508916159	5088.62883977
9	0	1273.166	2.03709655772	529.69096509	37	0	101.895	0.97569221824	15720.83878488
10	0	1324.292	0.74246356352	11506.76976979	38	0	84.711	3.67080093025	71430.69561813
11	0	901.855	2.04505443513	26.29831980	39	0	73.547	4.67926565481	801.82093112
12	0	1199.167	1.10962944315	1577.34354245	40	0	73.874	3.50319443167	3154.68708490
13	0	857.223	3.50849156957	398.14900341	41	0	78.756	3.03698313141	12036.46073489
14	0	779.786	1.17882652114	5223.69391980	42	0	79.637	1.80791330700	17260.15465469
15	0	990.250	5.23268129594	5884.92684658	43	0	85.803	5.98322631256	161000.68573767
16	0	753.141	2.53339053818	5507.55323867	44	0	56.963	2.78430398043	6286.59896834
17	0	505.264	4.58292563052	18849.22754997	45	0	61.148	1.81839811024	7084.89678112
18	0	492.379	4.20506639861	775.52261132	46	0	69.627	0.83297596966	9437.76293489
19	0	356.655	2.91954116867	0.06731030	47	0	56.116	4.38694880779	14143.49524243
20	0	284.125	1.89869034186	796.29800682	48	0	62.449	3.97763880587	8827.39026987
21	0	242.810	0.34481140906	5486.77784318	49	0	51.145	0.28306864501	5856.47765912
22	0	317.087	5.84901952218	11790.62908866	50	0	55.577	3.47006009062	6279.55273164
23	0	271.039	0.31488607649	10977.07880470	51	0	51.605	1.33282746983	1748.01641307
24	0	206.160	4.80646606059	2544.31441988	52	0	51.992	0.18914945834	12139.55350911
25	0	205.385	1.86947813692	5573.14280143	53	1628331966747.491	0.00000000000	0.00000000	
26	0	202.261	2.45767795458	6069.77675455	54	1	206058.863	2.67823455584	6283.07584999
27	0	126.184	1.08302630210	20.77539549	55	1	4303.430	2.63512650414	12566.15169998
28	0	155.516	0.83306073807	213.29909544	56	2	52918.870	0.00000000000	0.00000000

Matrix VB — latitude B (3 rows) and matrix VR — radius vector R (7 rows, A in 10^{-8} au)

Row	k	A	B	C	Row	k	A	B	C
1	0	279.620	3.19870156017	84334.66158131	1	0	100013988.799	0.00000000000	0.00000000
2	0	101.643	5.42248619256	5507.55323867	2	0	1670699.626	3.09846350771	6283.07584999
3	0	80.445	3.88013204458	5223.69391980	3	0	13956.023	3.05524609620	12566.15169998
					4	0	3083.720	5.19846674381	77713.77146812
					5	0	1628.461	1.17387749012	5753.38488490
					6	0	1575.568	2.84685245825	7860.41939244
					7	1	103018.608	1.10748969588	6283.07584999

Matrix NU — nutation, 10 largest terms (IAU 1980, Meeus Table 22.A)

Columns: multipliers of D, M, M', F, Ω , then S, S' ($\Delta\psi$, 0.0001") and C, C' ($\Delta\epsilon$, 0.0001").

Row	D	M	M'	F	Ω	S	S'	C	C'
1	0	0	0	0	1	-171996.0	-174.2	92025.0	8.9
2	-2	0	0	2	2	-13187.0	-1.6	5736.0	-3.1
3	0	0	0	2	2	-2274.0	-0.2	977.0	-0.5
4	0	0	0	0	2	2062.0	0.2	-895.0	0.5
5	0	1	0	0	0	1426.0	-3.4	54.0	-0.1
6	0	0	1	0	0	712.0	0.1	-7.0	0.0
7	-2	1	0	2	2	-517.0	1.2	224.0	-0.6
8	0	0	0	2	1	-386.0	-0.4	200.0	0.0
9	0	0	1	2	2	-301.0	0.0	129.0	-0.1
10	-2	-1	0	2	2	217.0	-0.5	-95.0	0.3

Method B — Chebyshev tables

This is the same idea as the USNO Almanac for Computers. For each time block, GHA and Dec are given as a Chebyshev series in the normalised time $x = 2t/\text{span} - 1$, where t is the number of hours since the start of the block (0h UT1 on the date shown). The C47 evaluates the series with the Clenshaw recurrence.

Body	Block length	Coefficients	Registers
Sun	8 days (span = 192 h)	GHA 5, Dec 5; SD printed for the block	$n = 5 \rightarrow R01, 192 \rightarrow R02, \text{GHA} \rightarrow R10\text{--}R14, \text{Dec} \rightarrow R20\text{--}R24$
Aries	8 days (192 h)	GHA 5	as Sun (Dec registers not used)
Venus, Mars, Jupiter, Saturn	8 days (192 h)	GHA 5, Dec 5	as Sun
Moon	1 day (24 h)	GHA 6, Dec 6, HP 4 (arcmin)	$n = 6 \rightarrow R01, 24 \rightarrow R02, \text{GHA} \rightarrow R10\text{--}R15, \text{Dec} \rightarrow R20\text{--}R25, \text{HP} \rightarrow R30\text{--}R33$

Procedure

- Find the line for the body whose start date is the last one on or before the date of the sight.
- Store n in R01, the span in hours in R02, and the coefficients in the registers above (c0 first).
- Key $t = 24 \times (\text{days since the start date}) + \text{UT in decimal hours}$, then XEQ **ALM**. It returns $X = \text{GHA}$ (0–360°) and $Y = \text{Dec}$ (degrees, negative = S).
- Moon only: after ALM, XEQ **HP** gives the horizontal parallax in arcminutes. $SD \approx 0.2725 \times \text{HP}$.
- Use →HMS (or the D.MS display) to show degrees and minutes.

Worked example — Sun, 23 Sep 2026, 12:00:00 UT1

Block 2026-09-17, so $t = 6 \times 24 + 12 = 156$ h. Coefficients:

	c0	c1	c2	c3	c4
GHA	1621.68475	1440.35480	-0.00108	-0.00024	-0.00000
Dec	0.77751	-1.55320	-0.00149	0.00024	-0.00000

Result: $\text{GHA} = 1.90695^\circ = \mathbf{1^\circ\ 54.42'}$, $\text{Dec} = -0.19313^\circ = \mathbf{S\ 0^\circ\ 11.59'}$ (USNO: $1^\circ\ 54.41'$, $S\ 0^\circ\ 11.59'$).

Program ALM (with CHEB and HP)

001	LBL "ALM"	; X=t hours	022	RCL 01	043	0
002	2		023	STO 09	044	STO 04 ; b1
003	x		024	4	045	STO 05 ; b2
004	RCL÷ 02	; span	025	STO 01	046	LBL 11
005	1		026	30	047	RCL 03
006	-		027	STO 08	048	RCL× 04
007	STO 03	; x	028	XEQ "CHEB"	049	2
008	10		029	RCL 09	050	x
009	STO 08		030	STO 01	051	RCL- 05
010	XEQ "CHEB"	; GHA	031	R↓	052	RCL+ IND 06 ; new b1
011	360		032	RTN	053	RCL 04
012	MOD		033	LBL "CHEB" ; Clenshaw	054	STO 05
013	STO 40		034	RCL 08	055	R↓
014	20		035	RCL+ 01	056	STO 04
015	STO 08		036	1	057	DSE 06
016	XEQ "CHEB"	; Dec	037	-	058	GTO 11
017	STO 41		038	RCL 08	059	RCL 03
018	RCL 41		039	1000	060	RCL× 04
019	RCL 40	; X,Y	040	÷	061	RCL- 05
020	RTN		041	+	062	RCL+ IND 08 ; result
021	LBL "HP"	; Moon HP	042	STO 06 ; counter	063	RTN

Other registers used: R03 x · R04–R05 Clenshaw terms · R06 loop counter · R08 base register · R09 saved n · R40 GHA · R41 Dec.

Sun — 8-day blocks (span 192 h, n = 5)

Sun 0h UT1	SD´	GHA c0	GHA c1	GHA c2	GHA c3	GHA c4	Dec c0	Dec c1	Dec c2	Dec c3	Dec c4
2026-09-01	15.86	1620.28737	1440.33286	0.00358	-0.00019	0.00000	6.87495	-1.48059	-0.00763	0.00025	0.00000
2026-09-09	15.89	1620.97490	1440.35292	0.00146	-0.00018	0.00000	3.86251	-1.52926	-0.00453	0.00026	0.00000
2026-09-17	15.93	1621.68475	1440.35480	-0.00108	-0.00024	0.00000	0.77751	-1.55320	-0.00149	0.00024	0.00000
2026-09-25	15.96	1622.37617	1440.33403	-0.00417	-0.00027	0.00000	-2.33178	-1.55376	0.00132	0.00024	0.00000
2026-10-03	16.00	1623.00151	1440.28901	-0.00694	-0.00019	0.00000	-5.41894	-1.53071	0.00453	0.00029	0.00000
2026-10-11	16.04	1623.51726	1440.22500	-0.00904	-0.00018	0.00000	-8.43313	-1.48057	0.00801	0.00029	0.00000
2026-10-19	16.07	1623.88811	1440.14404	-0.01122	-0.00019	0.00000	-11.31926	-1.40272	0.01142	0.00027	0.00000
2026-10-27	16.11	1624.07925	1440.04529	-0.01337	-0.00013	0.00001	-14.02293	-1.29816	0.01479	0.00031	0.00000
2026-11-04	16.14	1624.05954	1439.93438	-0.01412	0.00000	0.00001	-16.48912	-1.16492	0.01852	0.00031	0.00000
2026-11-12	16.17	1623.81617	1439.82260	-0.01371	0.00006	0.00000	-18.65928	-1.00225	0.02211	0.00028	0.00000
2026-11-20	16.20	1623.35445	1439.71645	-0.01278	0.00009	0.00000	-20.47672	-0.81262	0.02520	0.00024	0.00000
2026-11-28	16.22	1622.69015	1439.62087	-0.01082	0.00025	0.00001	-21.89130	-0.59961	0.02801	0.00021	-0.00001
2026-12-06	16.24	1621.85671	1439.54889	-0.00698	0.00037	0.00000	-22.85930	-0.36665	0.03010	0.00014	0.00000
2026-12-14	16.25	1620.91292	1439.51110	-0.00246	0.00037	0.00000	-23.34772	-0.12085	0.03119	0.00004	-0.00001
2026-12-22	16.26	1619.92861	1439.50794	0.00158	0.00032	0.00000	-23.33936	0.12923	0.03124	-0.00002	0.00000
2026-12-30	16.26	1618.97011	1439.53709	0.00579	0.00036	0.00000	-22.83298	0.37642	0.03037	-0.00013	-0.00001
2027-01-07	16.26	1618.10376	1439.59990	0.00978	0.00029	-0.00001	-21.84299	0.61190	0.02837	-0.00020	0.00000
2027-01-15	16.26	1617.39106	1439.68953	0.01233	0.00012	-0.00001	-20.40064	0.82812	0.02559	-0.00025	0.00000
2027-01-23	16.24	1616.87179	1439.79182	0.01310	0.00004	0.00000	-18.54916	1.02086	0.02257	-0.00027	-0.00001
2027-01-31	16.23	1616.56142	1439.89805	0.01339	0.00000	0.00000	-16.33804	1.18726	0.01896	-0.00032	0.00000
2027-02-08	16.20	1616.46371	1440.00385	0.01291	-0.00009	-0.00001	-13.82392	1.32367	0.01513	-0.00032	0.00000
2027-02-16	16.18	1616.56992	1440.10085	0.01111	-0.00021	0.00000	-11.06744	1.42976	0.01148	-0.00027	0.00000
2027-02-24	16.15	1616.85247	1440.17962	0.00862	-0.00019	0.00000	-8.12664	1.50820	0.00807	-0.00030	0.00000
2027-03-04	16.12	1617.27341	1440.23945	0.00636	-0.00018	0.00000	-5.05717	1.55825	0.00444	-0.00030	0.00000
2027-03-12	16.08	1617.79597	1440.28115	0.00399	-0.00023	-0.00001	-1.91639	1.57961	0.00093	-0.00028	0.00000
2027-03-20	16.05	1618.38070	1440.30099	0.00090	-0.00026	0.00000	1.24050	1.57479	-0.00207	-0.00024	0.00000
2027-03-28	16.01	1618.98061	1440.29659	-0.00190	-0.00020	0.00000	4.36382	1.54592	-0.00519	-0.00027	0.00000
2027-04-05	15.97	1619.55156	1440.27261	-0.00400	-0.00015	0.00000	7.40386	1.49141	-0.00843	-0.00027	0.00000
2027-04-13	15.94	1620.05876	1440.23292	-0.00601	-0.00019	0.00000	10.30932	1.41151	-0.01147	-0.00023	0.00000
2027-04-21	15.90	1620.46941	1440.17594	-0.00815	-0.00014	0.00001	13.03206	1.30896	-0.01420	-0.00024	0.00000
2027-04-29	15.87	1620.75174	1440.10540	-0.00931	-0.00004	0.00001	15.52705	1.18358	-0.01714	-0.00024	0.00000
2027-05-07	15.84	1620.88777	1440.03072	-0.00923	0.00003	0.00000	17.74804	1.03506	-0.01996	-0.00022	0.00000
2027-05-15	15.81	1620.87599	1439.95764	-0.00904	0.00003	0.00000	19.65072	0.86569	-0.02229	-0.00018	0.00000
2027-05-23	15.79	1620.72098	1439.88806	-0.00818	0.00012	0.00001	21.19701	0.67885	-0.02440	-0.00017	0.00000
2027-05-31	15.77	1620.43778	1439.83062	-0.00593	0.00026	0.00001	22.35371	0.47643	-0.02611	-0.00012	0.00000
2027-06-08	15.75	1620.06165	1439.79587	-0.00282	0.00023	0.00000	23.09389	0.26286	-0.02717	-0.00005	0.00000
2027-06-16	15.74	1619.63937	1439.78408	-0.00014	0.00023	0.00000	23.40094	0.04392	-0.02750	-0.00001	0.00000
2027-06-24	15.73	1619.21527	1439.79418	0.00271	0.00025	0.00000	23.26895	-0.17576	-0.02732	0.00005	0.00001
2027-07-02	15.73	1618.83559	1439.82893	0.00602	0.00027	-0.00001	22.70171	-0.39062	-0.02629	0.00011	0.00000
2027-07-10	15.73	1618.54984	1439.88720	0.00829	0.00013	-0.00001	21.71501	-0.59472	-0.02466	0.00015	0.00000
2027-07-18	15.74	1618.39440	1439.95823	0.00934	0.00006	0.00000	20.33420	-0.78451	-0.02276	0.00017	0.00000
2027-07-26	15.75	1618.38730	1440.03510	0.00987	0.00005	0.00000	18.59008	-0.95768	-0.02043	0.00022	0.00000
2027-08-03	15.77	1618.53798	1440.11584	0.01016	-0.00004	-0.00001	16.51998	-1.11011	-0.01766	0.00023	0.00000
2027-08-11	15.79	1618.84803	1440.19327	0.00903	-0.00014	0.00000	14.16719	-1.24042	-0.01495	0.00022	0.00000
2027-08-19	15.81	1619.30108	1440.25818	0.00713	-0.00017	0.00000	11.57495	-1.34965	-0.01235	0.00023	0.00000
2027-08-27	15.84	1619.86842	1440.30764	0.00531	-0.00013	0.00000	8.78604	-1.43678	-0.00937	0.00026	0.00000
2027-09-04	15.87	1620.52036	1440.34260	0.00328	-0.00021	0.00000	5.84719	-1.49956	-0.00635	0.00024	0.00000
2027-09-12	15.91	1621.22313	1440.35781	0.00045	-0.00025	0.00000	2.80615	-1.53918	-0.00359	0.00022	0.00000

Sun 0h UT1	SD'	GHA c0	GHA c1	GHA c2	GHA c3	GHA c4	Dec c0	Dec c1	Dec c2	Dec c3	Dec c4
2027-09-20	15.94	1621.93267	1440.34923	-0.00254	-0.00023	0.00001	-0.29213	-1.55672	-0.00072	0.00026	0.00000
2027-09-28	15.98	1622.60306	1440.31923	-0.00488	-0.00019	0.00000	-3.40095	-1.54936	0.00258	0.00028	0.00000
2027-10-06	16.01	1623.19486	1440.27048	-0.00738	-0.00023	0.00000	-6.46865	-1.51563	0.00583	0.00026	0.00000
2027-10-14	16.05	1623.66797	1440.20031	-0.01017	-0.00023	0.00000	-9.44335	-1.45646	0.00897	0.00027	0.00000
2027-10-22	16.08	1623.97967	1440.10959	-0.01231	-0.00012	0.00001	-12.27350	-1.37073	0.01253	0.00031	0.00000
2027-10-30	16.12	1624.09692	1440.00687	-0.01328	-0.00007	0.00000	-14.90293	-1.25560	0.01625	0.00031	0.00000
2027-11-07	16.15	1624.00216	1439.89782	-0.01394	-0.00004	0.00000	-17.27283	-1.11138	0.01976	0.00028	0.00000
2027-11-15	16.18	1623.68539	1439.78531	-0.01403	0.00005	0.00001	-19.32702	-0.94002	0.02310	0.00029	0.00000
2027-11-23	16.21	1623.14812	1439.67890	-0.01226	0.00023	0.00001	-21.01178	-0.74206	0.02630	0.00024	0.00000
2027-12-01	16.23	1622.41751	1439.59312	-0.00910	0.00029	0.00000	-22.27693	-0.52095	0.02886	0.00018	-0.00001
2027-12-09	16.24	1621.54207	1439.53437	-0.00558	0.00029	0.00000	-23.08222	-0.28297	0.03050	0.00010	0.00000
2027-12-17	16.26	1620.57812	1439.50499	-0.00159	0.00039	0.00001	-23.40082	-0.03488	0.03139	0.00002	-0.00001
2027-12-25	16.26	1619.59092	1439.51196	0.00336	0.00041	0.00000	-23.21984	0.21559	0.03107	-0.00007	-0.00001

Aries — GHA ♈, 8-day blocks (span 192 h, n = 5)

Aries 0h UT1	GHA c0	GHA c1	GHA c2	GHA c3	GHA c4
2026-09-01	1784.11810	1443.94266	0.00001	-0.00002	0.00000
2026-09-09	1792.00323	1443.94249	0.00002	0.00001	0.00000
2026-09-17	1799.88836	1443.94259	-0.00002	-0.00001	0.00000
2026-09-25	1447.77343	1443.94255	0.00005	0.00000	0.00000
2026-10-03	1455.65861	1443.94253	-0.00004	0.00001	0.00000
2026-10-11	1463.54370	1443.94262	0.00002	-0.00001	0.00000
2026-10-19	1471.42885	1443.94250	-0.00001	0.00001	0.00000
2026-10-27	1479.31404	1443.94268	-0.00002	-0.00002	0.00000
2026-11-04	1487.19922	1443.94256	0.00003	0.00000	0.00000
2026-11-12	1495.08446	1443.94262	-0.00003	0.00000	0.00000
2026-11-20	1502.96968	1443.94268	0.00005	-0.00001	-0.00001
2026-11-28	1510.85500	1443.94257	-0.00002	0.00001	0.00000
2026-12-06	1518.74026	1443.94270	0.00000	-0.00001	0.00000
2026-12-14	1526.62553	1443.94258	0.00002	0.00001	0.00000
2026-12-22	1534.51089	1443.94271	-0.00005	-0.00001	0.00001
2026-12-30	1542.39617	1443.94265	0.00003	0.00000	0.00000
2027-01-07	1550.28147	1443.94259	-0.00003	0.00000	0.00000
2027-01-15	1558.16672	1443.94272	0.00003	-0.00001	0.00000
2027-01-23	1566.05200	1443.94255	0.00001	0.00001	0.00000
2027-01-31	1573.93723	1443.94265	-0.00002	-0.00001	0.00000
2027-02-08	1581.82237	1443.94255	0.00004	0.00001	0.00000
2027-02-16	1589.70759	1443.94259	-0.00005	0.00000	0.00001
2027-02-24	1597.59272	1443.94261	0.00002	-0.00001	0.00000
2027-03-04	1605.47785	1443.94248	-0.00002	0.00001	0.00000
2027-03-12	1613.36296	1443.94265	0.00000	-0.00002	0.00000
2027-03-20	1621.24808	1443.94251	0.00003	0.00001	0.00000
2027-03-28	1629.13321	1443.94257	-0.00003	0.00000	0.00000
2027-04-05	1637.01828	1443.94258	0.00005	0.00000	0.00000
2027-04-13	1644.90347	1443.94253	-0.00003	0.00001	0.00000
2027-04-21	1652.78861	1443.94264	0.00001	-0.00001	0.00000
2027-04-29	1660.67377	1443.94252	0.00001	0.00001	0.00000
2027-05-07	1668.55901	1443.94268	-0.00003	-0.00001	0.00000
2027-05-15	1676.44421	1443.94261	0.00004	0.00000	0.00000
2027-05-23	1684.32948	1443.94260	-0.00003	0.00000	0.00000
2027-05-31	1692.21470	1443.94270	0.00004	-0.00001	0.00000
2027-06-08	1700.10002	1443.94257	0.00000	0.00002	0.00000
2027-06-16	1707.98530	1443.94269	-0.00002	-0.00001	0.00000
2027-06-24	1715.87053	1443.94258	0.00003	0.00001	0.00000
2027-07-02	1723.75586	1443.94267	-0.00005	0.00000	0.00001
2027-07-10	1731.64111	1443.94266	0.00003	-0.00001	0.00000
2027-07-18	1739.52637	1443.94255	-0.00002	0.00001	0.00000
2027-07-26	1747.41158	1443.94270	0.00002	-0.00002	0.00000
2027-08-03	1755.29682	1443.94254	0.00002	0.00001	0.00000

Aries 0h UT1	GHA c0	GHA c1	GHA c2	GHA c3	GHA c4
2027-08-11	1763.18202	1443.94261	-0.00003	0.00000	0.00000
2027-08-19	1771.06712	1443.94256	0.00004	0.00001	0.00000
2027-08-27	1778.95232	1443.94255	-0.00005	0.00001	0.00001
2027-09-04	1786.83744	1443.94262	0.00001	-0.00001	0.00000
2027-09-12	1794.72254	1443.94247	0.00000	0.00001	0.00000
2027-09-20	1442.60765	1443.94264	-0.00001	-0.00002	0.00000
2027-09-28	1450.49276	1443.94253	0.00004	0.00000	0.00000
2027-10-06	1458.37791	1443.94254	-0.00003	0.00000	0.00000
2027-10-14	1466.26298	1443.94261	0.00004	0.00000	0.00000
2027-10-22	1474.14817	1443.94252	-0.00002	0.00002	0.00000
2027-10-30	1482.03334	1443.94266	-0.00001	-0.00001	0.00000
2027-11-07	1489.91851	1443.94254	0.00002	0.00001	0.00000
2027-11-15	1497.80377	1443.94266	-0.00004	-0.00001	0.00000
2027-11-23	1505.68900	1443.94265	0.00004	-0.00001	0.00000
2027-12-01	1513.57429	1443.94258	-0.00003	0.00001	0.00000
2027-12-09	1521.45955	1443.94273	0.00003	-0.00001	0.00000
2027-12-17	1529.34487	1443.94258	0.00001	0.00001	0.00000
2027-12-25	1537.23018	1443.94269	-0.00003	-0.00001	0.00000

Venus — 8-day blocks (span 192 h, n = 5)

Venus 0h UT1	GHA c0	GHA c1	GHA c2	GHA c3	GHA c4	Dec c0	Dec c1	Dec c2	Dec c3	Dec c4
2026-09-01	1581.27204	1441.30275	0.05015	0.00144	0.00005	-13.74420	-1.57033	0.02465	0.00062	0.00001
2026-09-09	1584.34346	1441.78731	0.07252	0.00229	0.00005	-16.66121	-1.33922	0.03360	0.00089	0.00002
2026-09-17	1588.59318	1442.48845	0.10369	0.00281	0.00001	-19.03266	-1.02146	0.04655	0.00129	0.00003
2026-09-25	1594.50326	1443.44788	0.13498	0.00212	-0.00010	-20.64833	-0.57893	0.06488	0.00175	0.00002
2026-10-03	1602.53268	1444.59077	0.14572	-0.00077	-0.00028	-21.21802	0.02759	0.08673	0.00170	-0.00006
2026-10-11	1612.78740	1445.62904	0.10358	-0.00656	-0.00039	-20.42441	0.77372	0.09557	-0.00073	-0.00025
2026-10-19	1624.56308	1446.05693	-0.00331	-0.01030	-0.00001	-18.19347	1.42751	0.05995	-0.00517	-0.00023
2026-10-27	1636.28411	1445.57405	-0.11050	-0.00646	0.00041	-15.08227	1.62271	-0.01247	-0.00592	0.00014
2026-11-04	1646.38045	1444.48949	-0.15017	-0.00054	0.00027	-12.12557	1.29045	-0.06457	-0.00255	0.00022
2026-11-12	1654.17831	1443.31888	-0.13795	0.00207	0.00007	-10.11992	0.70507	-0.07694	0.00020	0.00011
2026-11-20	1659.79850	1442.32458	-0.11018	0.00236	-0.00001	-9.30095	0.12242	-0.06687	0.00129	0.00004
2026-11-28	1663.65379	1441.55327	-0.08317	0.00205	-0.00004	-9.53686	-0.34354	-0.04922	0.00153	-0.00001
2026-12-06	1666.16482	1440.97489	-0.06263	0.00137	-0.00004	-10.56175	-0.66699	-0.03200	0.00133	-0.00001
2026-12-14	1667.65817	1440.52914	-0.04980	0.00080	-0.00003	-12.10412	-0.86327	-0.01747	0.00110	-0.00001
2026-12-22	1668.34392	1440.16289	-0.04220	0.00055	0.00000	-13.93016	-0.95236	-0.00495	0.00101	-0.00001
2026-12-30	1668.35257	1439.85102	-0.03590	0.00048	-0.00001	-15.83772	-0.94579	0.00633	0.00087	-0.00001
2027-01-07	1667.78446	1439.58520	-0.03070	0.00040	0.00000	-17.64726	-0.85574	0.01595	0.00073	-0.00001
2027-01-15	1666.72375	1439.35784	-0.02619	0.00037	0.00000	-19.20490	-0.69522	0.02397	0.00061	0.00000
2027-01-23	1665.24556	1439.16842	-0.02083	0.00055	0.00001	-20.38121	-0.47540	0.03077	0.00049	-0.00001
2027-01-31	1663.43802	1439.03006	-0.01362	0.00063	0.00000	-21.06939	-0.20878	0.03558	0.00031	-0.00001
2027-02-08	1661.41319	1438.95139	-0.00611	0.00060	-0.00001	-21.19267	0.08770	0.03821	0.00013	-0.00001
2027-02-16	1659.28876	1438.92971	0.00054	0.00052	0.00000	-20.70870	0.39676	0.03881	-0.00002	-0.00001
2027-02-24	1657.17181	1438.95834	0.00653	0.00046	-0.00001	-19.60780	0.70295	0.03739	-0.00021	-0.00001
2027-03-04	1655.15643	1439.03009	0.01110	0.00029	-0.00001	-17.91235	0.98974	0.03409	-0.00033	-0.00001
2027-03-12	1653.31358	1439.12878	0.01317	0.00006	-0.00001	-15.67400	1.24483	0.02956	-0.00041	0.00000
2027-03-20	1651.67630	1439.23359	0.01277	-0.00010	-0.00001	-12.96352	1.46146	0.02452	-0.00045	-0.00001
2027-03-28	1650.24082	1439.32949	0.01104	-0.00019	-0.00001	-9.86239	1.63487	0.01877	-0.00049	0.00000
2027-04-05	1648.97957	1439.40684	0.00813	-0.00029	-0.00001	-6.46141	1.76113	0.01279	-0.00050	0.00000
2027-04-13	1647.84573	1439.45581	0.00393	0.00040	0.00000	-2.85567	1.83972	0.00692	-0.00047	0.00000
2027-04-21	1646.77335	1439.46774	-0.00095	-0.00040	0.00000	0.86094	1.87204	0.00110	-0.00050	0.00000
2027-04-29	1645.68618	1439.44117	-0.00565	-0.00038	0.00000	4.59454	1.85647	-0.00501	-0.00051	0.00000
2027-05-07	1644.50942	1439.37844	-0.01001	-0.00036	0.00000	8.24785	1.79167	-0.01121	-0.00052	0.00000
2027-05-15	1643.17256	1439.28117	-0.01423	-0.00032	0.00001	11.72220	1.67762	-0.01730	-0.00051	0.00000
2027-05-23	1641.61031	1439.15401	-0.01732	-0.00019	0.00001	14.91917	1.51408	-0.02362	-0.00053	0.00000
2027-05-31	1639.77492	1439.00968	-0.01841	0.00001	0.00001	17.73831	1.29983	-0.02990	-0.00051	0.00000
2027-06-08	1637.64927	1438.86553	-0.01738	0.00016	0.00001	20.07989	1.03690	-0.03571	-0.00044	0.00001
2027-06-16	1635.24946	1438.73718	-0.01435	0.00035	0.00001	21.85201	0.73116	-0.04059	-0.00037	0.00001
2027-06-24	1632.62479	1438.64267	-0.00894	0.00054	0.00001	22.97716	0.39095	-0.04422	-0.00023	0.00001
2027-07-02	1629.86106	1438.59974	-0.00160	0.00064	0.00000	23.39877	0.02925	-0.04591	-0.00006	0.00001
2027-07-10	1627.07139	1438.61667	0.00570	0.00057	-0.00001	23.09000	-0.33767	-0.04549	0.00012	0.00001
2027-07-18	1624.37050	1438.68753	0.01179	0.00044	-0.00001	22.05707	-0.69333	-0.04317	0.00027	0.00001
2027-07-26	1621.85455	1438.80009	0.01608	0.00028	-0.00001	20.33694	-1.02340	-0.03909	0.00041	0.00001
2027-08-03	1619.59169	1438.93884	0.01820	0.00005	-0.00002	17.99398	-1.31508	-0.03370	0.00048	0.00000
2027-08-11	1617.61443	1439.08338	0.01757	-0.00015	-0.00001	15.11304	-1.56084	-0.02768	0.00051	0.00000
2027-08-19	1615.91410	1439.21400	0.01484	-0.00029	-0.00001	11.78956	-1.75745	-0.02145	0.00053	0.00000
2027-08-27	1614.44881	1439.31745	0.01093	-0.00035	0.00000	8.12368	-1.90296	-0.01490	0.00055	0.00000
2027-09-04	1613.15665	1439.38642	0.00616	-0.00044	0.00000	4.21935	-1.99592	-0.00835	0.00054	0.00000
2027-09-12	1611.96132	1439.41357	0.00055	-0.00049	0.00000	0.18101	-2.03709	-0.00196	0.00053	0.00000

Venus 0h UT1	GHA c0	GHA c1	GHA c2	GHA c3	GHA c4	Dec c0	Dec c1	Dec c2	Dec c3	Dec c4
2027-09-20	1610.77424	1439.39446	-0.00530	-0.00047	0.00000	-3.88867	-2.02722	0.00449	0.00056	0.00000
2027-09-28	1609.50358	1439.33051	-0.01056	-0.00041	0.00000	-7.88572	-1.96417	0.01129	0.00057	0.00000
2027-10-06	1608.06510	1439.22715	-0.01519	-0.00036	0.00000	-11.70189	-1.84624	0.01821	0.00058	0.00000
2027-10-14	1606.38531	1439.08994	-0.01893	-0.00025	0.00001	-15.22683	-1.67295	0.02512	0.00058	0.00000
2027-10-22	1604.40649	1438.92971	-0.02074	-0.00003	0.00002	-18.34953	-1.44388	0.03216	0.00058	0.00000
2027-10-30	1602.10207	1438.76691	-0.01952	0.00022	0.00001	-20.95867	-1.15974	0.03878	0.00052	-0.00001
2027-11-07	1599.49089	1438.62525	-0.01551	0.00044	0.00001	-22.94938	-0.82631	0.04437	0.00040	-0.00001
2027-11-15	1596.63641	1438.52563	-0.00907	0.00063	0.00001	-24.23357	-0.45460	0.04831	0.00026	-0.00001
2027-11-23	1593.64136	1438.48653	-0.00043	0.00079	0.00000	-24.74895	-0.05926	0.05013	0.00004	-0.00001
2027-12-01	1590.64098	1438.52091	0.00893	0.00074	-0.00001	-24.46735	0.34023	0.04938	-0.00017	-0.00001
2027-12-09	1587.78025	1438.62483	0.01668	0.00053	-0.00002	-23.40059	0.72390	0.04622	-0.00035	-0.00001
2027-12-17	1585.18061	1438.77964	0.02163	0.00031	-0.00001	-21.59753	1.07513	0.04137	-0.00047	-0.00001
2027-12-25	1582.92238	1438.96426	0.02418	0.00011	-0.00001	-19.13562	1.38151	0.03506	-0.00056	0.00000

Mars — 8-day blocks (span 192 h, n = 5)

Mars 0h UT1	GHA c0	GHA c1	GHA c2	GHA c3	GHA c4	Dec c0	Dec c1	Dec c2	Dec c3	Dec c4
2026-09-01	1676.78169	1441.22207	0.00887	0.00012	0.00000	23.06930	-0.23714	-0.01040	0.00010	0.00000
2026-09-09	1679.30127	1441.29862	0.01018	0.00008	0.00000	22.51567	-0.31547	-0.00916	0.00011	0.00000
2026-09-17	1681.98254	1441.38328	0.01093	0.00005	0.00000	21.81568	-0.38341	-0.00783	0.00011	0.00000
2026-09-25	1684.83837	1441.47305	0.01153	0.00007	0.00000	20.99024	-0.44096	-0.00653	0.00012	0.00000
2026-10-03	1687.87994	1441.56944	0.01262	0.00010	0.00000	20.06112	-0.48684	-0.00494	0.00013	0.00000
2026-10-11	1691.12293	1441.67428	0.01352	0.00006	0.00000	19.05306	-0.51987	-0.00331	0.00013	0.00000
2026-10-19	1694.58171	1441.78504	0.01416	0.00005	0.00000	17.99174	-0.54019	-0.00179	0.00012	0.00000
2026-10-27	1698.26790	1441.90203	0.01524	0.00014	0.00000	16.90219	-0.54790	-0.00005	0.00016	0.00000
2026-11-04	1702.19931	1442.03082	0.01696	0.00014	0.00000	15.81209	-0.54060	0.00188	0.00016	0.00000
2026-11-12	1706.40195	1442.17320	0.01864	0.00014	0.00000	14.75207	-0.51782	0.00379	0.00015	0.00000
2026-11-20	1710.90323	1442.32968	0.02058	0.00020	0.00001	13.75259	-0.48010	0.00568	0.00018	0.00001
2026-11-28	1715.73678	1442.50661	0.02385	0.00031	0.00000	12.84534	-0.42511	0.00810	0.00021	0.00000
2026-12-06	1720.95287	1442.71271	0.02771	0.00033	0.00000	12.06778	-0.35038	0.01058	0.00020	0.00000
2026-12-14	1726.61288	1442.95075	0.03185	0.00036	0.00000	11.45904	-0.25644	0.01288	0.00018	0.00000
2026-12-22	1732.78424	1443.22485	0.03695	0.00051	0.00001	11.05679	-0.14370	0.01541	0.00023	0.00000
2026-12-30	1739.54913	1443.54518	0.04309	0.00049	-0.00001	10.90082	-0.01023	0.01786	0.00017	-0.00001
2027-01-07	1747.00114	1443.91101	0.04810	0.00033	-0.00001	11.02821	0.13870	0.01913	0.00003	-0.00001
2027-01-15	1755.21774	1444.30773	0.05068	0.00010	-0.00001	11.45793	0.29055	0.01858	-0.00011	-0.00001
2027-01-23	1764.23953	1444.71367	0.05015	-0.00028	-0.00004	12.18127	0.43068	0.01602	-0.00035	-0.00002
2027-01-31	1774.04860	1445.08891	0.04234	-0.00104	-0.00005	13.15345	0.53634	0.00983	-0.00067	-0.00002
2027-02-08	1784.51682	1445.36530	0.02556	-0.00171	-0.00003	14.27639	0.57880	0.00050	-0.00083	0.00000
2027-02-16	1795.38345	1445.48297	0.00358	-0.00188	0.00000	15.40772	0.54483	-0.00873	-0.00069	0.00001
2027-02-24	1446.30619	1445.42090	-0.01902	-0.00183	0.00002	16.40351	0.44504	-0.01583	-0.00047	0.00002
2027-03-04	1456.93154	1445.18845	-0.03817	-0.00130	0.00004	17.15261	0.30083	-0.01970	-0.00017	0.00002
2027-03-12	1466.96302	1444.83398	-0.04901	-0.00049	0.00005	17.59398	0.14040	-0.01988	0.00012	0.00001
2027-03-20	1476.22848	1444.42977	-0.05113	0.00004	0.00002	17.72180	-0.01084	-0.01782	0.00019	0.00000
2027-03-28	1484.68398	1444.02755	-0.04896	0.00031	0.00001	17.56480	-0.14430	-0.01559	0.00017	0.00000
2027-04-05	1492.36131	1443.65379	-0.04415	0.00050	0.00001	17.15780	-0.26105	-0.01360	0.00017	0.00000
2027-04-13	1499.33649	1443.32706	-0.03738	0.00058	-0.00001	16.53344	-0.36162	-0.01159	0.00014	0.00000
2027-04-21	1505.71235	1443.05402	-0.03110	0.00046	-0.00001	15.72228	-0.44838	-0.01019	0.00009	0.00000
2027-04-29	1511.58810	1442.82594	-0.02605	0.00039	0.00000	14.74685	-0.52636	-0.00936	0.00006	0.00000
2027-05-07	1517.04608	1442.63589	-0.02144	0.00039	0.00000	13.62180	-0.59794	-0.00847	0.00008	0.00000
2027-05-15	1522.16033	1442.48184	-0.01727	0.00029	-0.00001	12.36104	-0.66211	-0.00761	0.00006	0.00000
2027-05-23	1526.99545	1442.35562	-0.01445	0.00019	0.00000	10.97796	-0.72048	-0.00703	0.00004	0.00000
2027-05-31	1531.59773	1442.24833	-0.01243	0.00016	0.00000	9.48217	-0.77492	-0.00654	0.00006	0.00000
2027-06-08	1536.00145	1442.15711	-0.01039	0.00015	-0.00001	7.88255	-0.82402	-0.00573	0.00006	0.00000
2027-06-16	1540.23739	1442.07996	-0.00904	0.00008	0.00000	6.19096	-0.86698	-0.00503	0.00005	0.00000
2027-06-24	1544.32726	1442.01041	-0.00843	0.00003	0.00000	4.41855	-0.90499	-0.00449	0.00004	0.00000
2027-07-02	1548.28184	1441.94454	-0.00798	0.00006	0.00000	2.57472	-0.93820	-0.00373	0.00008	0.00000
2027-07-10	1552.10895	1441.88301	-0.00747	0.00002	0.00000	0.67130	-0.96449	-0.00285	0.00007	0.00000
2027-07-18	1555.81525	1441.82323	-0.00755	-0.00003	0.00000	-1.27789	-0.98404	-0.00206	0.00006	0.00000
2027-07-26	1559.39998	1441.76113	-0.00799	-0.00004	0.00000	-3.26020	-0.99765	-0.00130	0.00008	0.00000
2027-08-03	1562.85748	1441.69622	-0.00818	-0.00001	0.00000	-5.26255	-1.00381	-0.00023	0.00009	0.00000
2027-08-11	1566.18371	1441.62972	-0.00851	-0.00005	0.00000	-7.26858	-1.00130	0.00086	0.00009	0.00000
2027-08-19	1569.37289	1441.55885	-0.00925	-0.00007	0.00000	-9.26110	-0.99039	0.00186	0.00008	0.00000
2027-08-27	1572.41411	1441.48175	-0.00997	-0.00004	0.00000	-11.22336	-0.97085	0.00309	0.00011	0.00000
2027-09-04	1575.29675	1441.40061	-0.01031	-0.00003	0.00000	-13.13605	-0.94070	0.00446	0.00012	0.00000
2027-09-12	1578.01394	1441.31613	-0.01085	-0.00005	0.00000	-14.97730	-0.89939	0.00585	0.00011	0.00000

Mars 0h UT1	GHA c0	GHA c1	GHA c2	GHA c3	GHA c4	Dec c0	Dec c1	Dec c2	Dec c3	Dec c4
2027-09-20	1580.55730	1441.22668	-0.01150	-0.00005	0.00000	-16.72497	-0.84713	0.00725	0.00013	0.00000
2027-09-28	1582.91767	1441.13355	-0.01167	0.00002	0.00000	-18.35610	-0.78265	0.00886	0.00013	0.00000
2027-10-06	1585.09211	1441.04108	-0.01145	0.00002	0.00000	-19.84533	-0.70522	0.01050	0.00014	0.00000
2027-10-14	1587.08329	1440.95027	-0.01125	0.00002	0.00000	-21.16683	-0.61501	0.01203	0.00012	0.00000
2027-10-22	1588.89512	1440.86200	-0.01072	0.00008	0.00001	-22.29568	-0.51252	0.01361	0.00013	0.00000
2027-10-30	1590.53719	1440.78120	-0.00938	0.00013	0.00000	-23.20723	-0.39785	0.01504	0.00012	0.00000
2027-11-07	1592.02955	1440.71249	-0.00779	0.00013	0.00000	-23.87849	-0.27236	0.01629	0.00009	0.00000
2027-11-15	1593.39735	1440.65668	-0.00614	0.00015	0.00000	-24.28984	-0.13819	0.01725	0.00008	0.00000
2027-11-23	1594.66782	1440.61555	-0.00403	0.00020	0.00000	-24.42574	0.00286	0.01793	0.00003	0.00000
2027-12-01	1595.87457	1440.59328	-0.00155	0.00020	0.00000	-24.27551	0.14760	0.01821	0.00001	0.00000
2027-12-09	1597.05592	1440.58990	0.00064	0.00016	0.00000	-23.83482	0.29297	0.01807	-0.00003	0.00000
2027-12-17	1598.24668	1440.60239	0.00247	0.00016	0.00000	-23.10548	0.43604	0.01765	-0.00006	0.00000
2027-12-25	1599.47727	1440.62979	0.00437	0.00015	0.00000	-22.09487	0.57380	0.01673	-0.00009	0.00000

Jupiter — 8-day blocks (span 192 h, n = 5)

Jupiter 0h UT1	GHA c0	GHA c1	GHA c2	GHA c3	GHA c4	Dec c0	Dec c1	Dec c2	Dec c3	Dec c4
2026-09-01	1646.95852	1443.11132	0.00352	0.00006	0.00000	16.98515	-0.23394	0.00015	0.00003	0.00000
2026-09-09	1653.21182	1443.14262	0.00430	0.00006	0.00000	16.51963	-0.23130	0.00049	0.00003	0.00000
2026-09-17	1659.53362	1443.17975	0.00498	0.00006	0.00000	16.06216	-0.22584	0.00087	0.00003	0.00000
2026-09-25	1665.93506	1443.22223	0.00564	0.00006	0.00000	15.61841	-0.21766	0.00118	0.00003	0.00000
2026-10-03	1672.42723	1443.27066	0.00650	0.00008	0.00000	15.19402	-0.20634	0.00164	0.00003	0.00000
2026-10-11	1679.02331	1443.32613	0.00735	0.00007	0.00000	14.79573	-0.19158	0.00207	0.00004	0.00000
2026-10-19	1685.73695	1443.38816	0.00815	0.00006	0.00000	14.43049	-0.17334	0.00246	0.00003	0.00000
2026-10-27	1692.58098	1443.45658	0.00900	0.00008	0.00000	14.10479	-0.15198	0.00292	0.00004	0.00000
2026-11-04	1699.56926	1443.53251	0.00997	0.00007	0.00000	13.82557	-0.12688	0.00335	0.00004	0.00000
2026-11-12	1706.71670	1443.61561	0.01078	0.00006	0.00000	13.59998	-0.09837	0.00376	0.00003	0.00000
2026-11-20	1714.03635	1443.70460	0.01145	0.00006	0.00000	13.43416	-0.06723	0.00403	0.00003	0.00000
2026-11-28	1721.53945	1443.79911	0.01217	0.00005	0.00000	13.33311	-0.03355	0.00435	0.00002	0.00000
2026-12-06	1729.23663	1443.89843	0.01260	0.00002	0.00000	13.30134	0.00193	0.00451	0.00001	0.00000
2026-12-14	1737.13454	1443.99948	0.01260	-0.00002	0.00000	13.34124	0.03790	0.00443	-0.00002	0.00000
2026-12-22	1745.23339	1444.09912	0.01229	-0.00003	0.00000	13.45199	0.07274	0.00428	-0.00002	0.00000
2026-12-30	1753.52815	1444.19506	0.01159	-0.00009	0.00000	13.63055	0.10549	0.00388	-0.00004	0.00000
2027-01-07	1762.00688	1444.28249	0.01017	-0.00015	0.00000	13.87078	0.13420	0.00324	-0.00007	0.00000
2027-01-15	1770.64713	1444.35607	0.00816	-0.00018	0.00000	14.16232	0.15660	0.00237	-0.00007	0.00000
2027-01-23	1779.41747	1444.41237	0.00586	-0.00021	0.00000	14.49170	0.17200	0.00143	-0.00009	0.00000
2027-01-31	1788.28033	1444.44812	0.00301	-0.00025	0.00000	14.84391	0.17935	0.00039	-0.00009	0.00000
2027-02-08	1797.19072	1444.45963	-0.00015	-0.00027	0.00000	15.20211	0.17789	-0.00075	-0.00009	0.00000
2027-02-16	1446.09909	1444.44626	-0.00312	-0.00023	0.00000	15.54886	0.16811	-0.00166	-0.00008	0.00000
2027-02-24	1454.95791	1444.41026	-0.00587	-0.00022	0.00000	15.86906	0.15141	-0.00247	-0.00006	0.00000
2027-03-04	1463.72335	1444.35309	-0.00837	-0.00019	0.00000	16.14993	0.12888	-0.00315	-0.00005	0.00000
2027-03-12	1472.35606	1444.27802	-0.01028	-0.00012	0.00001	16.38083	0.10167	-0.00358	-0.00002	0.00000
2027-03-20	1480.82587	1444.19082	-0.01145	-0.00008	0.00000	16.55476	0.07206	-0.00381	-0.00001	0.00000
2027-03-28	1489.11317	1444.09582	-0.01224	-0.00005	0.00000	16.66808	0.04119	-0.00390	-0.00001	0.00000
2027-04-05	1497.20553	1443.99626	-0.01258	0.00000	0.00000	16.71895	0.00963	-0.00395	0.00001	0.00000
2027-04-13	1505.09797	1443.89643	-0.01230	0.00004	0.00000	16.70711	-0.02135	-0.00381	0.00001	0.00000
2027-04-21	1512.79394	1443.79996	-0.01181	0.00005	0.00000	16.63442	-0.05120	-0.00364	0.00001	0.00000
2027-04-29	1520.30127	1443.70789	-0.01119	0.00006	0.00000	16.50328	-0.07984	-0.00353	0.00001	0.00000
2027-05-07	1527.62997	1443.62152	-0.01034	0.00009	0.00000	16.31595	-0.10729	-0.00330	0.00002	0.00000
2027-05-15	1534.79355	1443.54289	-0.00934	0.00008	0.00000	16.07556	-0.13294	-0.00311	0.00002	0.00000
2027-05-23	1541.80753	1443.47183	-0.00844	0.00007	0.00000	15.78552	-0.15694	-0.00292	0.00001	0.00000
2027-05-31	1548.68642	1443.40778	-0.00756	0.00008	0.00000	15.44869	-0.17977	-0.00276	0.00002	0.00000
2027-06-08	1555.44469	1443.35134	-0.00654	0.00008	0.00000	15.06791	-0.20082	-0.00253	0.00001	0.00000
2027-06-16	1562.09787	1443.30256	-0.00567	0.00007	0.00000	14.64669	-0.22022	-0.00232	0.00002	0.00000
2027-06-24	1568.66004	1443.26023	-0.00492	0.00006	0.00000	14.18817	-0.23820	-0.00219	0.00001	0.00000
2027-07-02	1575.14356	1443.22395	-0.00412	0.00007	0.00000	13.69501	-0.25475	-0.00194	0.00002	0.00000
2027-07-10	1581.56115	1443.19429	-0.00333	0.00006	0.00000	13.17063	-0.26946	-0.00173	0.00002	0.00000
2027-07-18	1587.92517	1443.17025	-0.00269	0.00005	0.00000	12.61858	-0.28244	-0.00154	0.00001	0.00000
2027-07-26	1594.24596	1443.15103	-0.00210	0.00005	0.00000	12.04189	-0.29408	-0.00134	0.00003	0.00000
2027-08-03	1600.53343	1443.13704	-0.00140	0.00005	0.00000	11.44392	-0.30369	-0.00109	0.00002	0.00000
2027-08-11	1606.79822	1443.12824	-0.00082	0.00004	0.00000	10.82868	-0.31132	-0.00083	0.00002	0.00000
2027-08-19	1613.04977	1443.12373	-0.00032	0.00004	0.00000	10.20003	-0.31719	-0.00064	0.00002	0.00000
2027-08-27	1619.29651	1443.12355	0.00030	0.00003	-0.00001	9.56152	-0.32105	-0.00036	0.00000	0.00001
2027-09-04	1625.54786	1443.12832	0.00090	0.00005	0.00000	8.91759	-0.32260	-0.00003	0.00003	0.00000
2027-09-12	1631.81344	1443.13768	0.00143	0.00004	0.00000	8.27300	-0.32177	0.00022	0.00002	0.00000

Jupiter 0h UT1	GHA c0	GHA c1	GHA c2	GHA c3	GHA c4	Dec c0	Dec c1	Dec c2	Dec c3	Dec c4
2027-09-20	1638.10188	1443.15122	0.00198	0.00005	0.00000	7.63208	-0.31890	0.00053	0.00003	0.00000
2027-09-28	1644.42233	1443.16982	0.00268	0.00006	0.00000	6.99966	-0.31324	0.00087	0.00003	0.00000
2027-10-06	1650.78547	1443.19385	0.00332	0.00005	0.00000	6.38135	-0.30476	0.00123	0.00002	0.00000
2027-10-14	1657.20163	1443.22282	0.00393	0.00005	0.00000	5.78259	-0.29376	0.00153	0.00003	0.00000
2027-10-22	1663.68101	1443.25720	0.00470	0.00007	0.00000	5.20868	-0.27980	0.00195	0.00003	0.00000
2027-10-30	1670.23566	1443.29814	0.00552	0.00006	0.00000	4.66594	-0.26259	0.00237	0.00004	0.00000
2027-11-07	1676.87852	1443.34535	0.00628	0.00006	0.00000	4.16085	-0.24221	0.00271	0.00003	0.00000
2027-11-15	1683.62188	1443.39869	0.00709	0.00008	0.00000	3.69936	-0.21893	0.00314	0.00004	0.00000
2027-11-23	1690.47901	1443.45925	0.00805	0.00008	0.00000	3.28796	-0.19213	0.00356	0.00004	0.00000
2027-12-01	1697.46473	1443.52718	0.00891	0.00007	0.00000	2.93354	-0.16195	0.00396	0.00003	0.00000
2027-12-09	1704.59284	1443.60156	0.00968	0.00006	0.00000	2.64226	-0.12908	0.00427	0.00003	0.00000
2027-12-17	1711.87599	1443.68230	0.01052	0.00007	0.00000	2.41958	-0.09329	0.00464	0.00002	0.00000
2027-12-25	1719.32716	1443.76946	0.01122	0.00005	0.00000	2.27108	-0.05496	0.00493	0.00002	0.00000

Saturn — 8-day blocks (span 192 h, n = 5)

Saturn 0h UT1	GHA c0	GHA c1	GHA c2	GHA c3	GHA c4	Dec c0	Dec c1	Dec c2	Dec c3	Dec c4
2026-09-01	1770.71419	1444.16278	0.00428	-0.00007	0.00000	2.85013	-0.10523	-0.00150	0.00003	0.00000
2026-09-09	1779.07107	1444.19328	0.00330	-0.00009	0.00000	2.62892	-0.11559	-0.00105	0.00005	0.00000
2026-09-17	1787.48051	1444.21521	0.00217	-0.00010	0.00000	2.39100	-0.12192	-0.00055	0.00004	0.00000
2026-09-25	1795.92467	1444.22799	0.00102	-0.00010	0.00000	2.14431	-0.12433	-0.00003	0.00005	0.00000
2026-10-03	1444.38499	1444.23130	-0.00022	-0.00011	0.00000	1.89696	-0.12261	0.00047	0.00005	0.00000
2026-10-11	1452.84165	1444.22425	-0.00153	-0.00011	0.00000	1.65746	-0.11640	0.00106	0.00004	0.00000
2026-10-19	1461.27401	1444.20712	-0.00272	-0.00009	0.00000	1.43457	-0.10610	0.00152	0.00004	0.00000
2026-10-27	1469.66313	1444.18114	-0.00376	-0.00009	0.00000	1.23598	-0.09215	0.00193	0.00003	0.00000
2026-11-04	1477.99208	1444.14697	-0.00476	-0.00008	0.00000	1.06839	-0.07508	0.00235	0.00003	0.00000
2026-11-12	1486.24527	1444.10556	-0.00555	-0.00005	0.00000	0.93793	-0.05517	0.00260	0.00002	0.00000
2026-11-20	1494.41017	1444.05892	-0.00607	-0.00003	0.00000	0.84905	-0.03353	0.00280	0.00001	0.00000
2026-11-28	1502.47816	1444.00874	-0.00647	-0.00003	0.00000	0.80461	-0.01086	0.00289	0.00001	0.00000
2026-12-06	1510.44303	1443.95593	-0.00670	-0.00001	0.00000	0.80633	0.01262	0.00294	-0.00001	0.00000
2026-12-14	1518.30115	1443.90219	-0.00670	0.00001	0.00000	0.85490	0.03589	0.00288	0.00000	0.00000
2026-12-22	1526.05244	1443.84924	-0.00653	0.00001	0.00000	0.94939	0.05848	0.00274	-0.00001	0.00000
2026-12-30	1533.69920	1443.79769	-0.00634	0.00002	0.00000	1.08796	0.08002	0.00264	-0.00001	0.00000
2027-01-07	1541.24488	1443.74828	-0.00599	0.00004	0.00000	1.26844	0.10027	0.00241	-0.00002	0.00000
2027-01-15	1548.69498	1443.70222	-0.00551	0.00004	0.00000	1.48764	0.11875	0.00220	-0.00003	0.00000
2027-01-23	1556.05681	1443.66000	-0.00506	0.00004	0.00000	1.74178	0.13519	0.00195	-0.00001	0.00000
2027-01-31	1563.33786	1443.62146	-0.00456	0.00004	0.00000	2.02710	0.14991	0.00170	-0.00003	0.00000
2027-02-08	1570.54608	1443.58723	-0.00398	0.00005	0.00000	2.33961	0.16238	0.00143	-0.00002	0.00000
2027-02-16	1577.69059	1443.55777	-0.00340	0.00004	0.00000	2.67491	0.17265	0.00113	-0.00002	0.00000
2027-02-24	1584.78055	1443.53263	-0.00288	0.00005	0.00000	3.02861	0.18088	0.00092	-0.00002	0.00000
2027-03-04	1591.82452	1443.51182	-0.00231	0.00005	0.00000	3.39676	0.18702	0.00062	-0.00002	0.00000
2027-03-12	1598.83160	1443.49578	-0.00170	0.00005	0.00000	3.77502	0.19103	0.00036	-0.00003	0.00000
2027-03-20	1605.81128	1443.48432	-0.00118	0.00004	0.00000	4.15913	0.19289	0.00014	-0.00001	0.00000
2027-03-28	1612.77212	1443.47696	-0.00066	0.00005	0.00000	4.54526	0.19303	-0.00012	-0.00002	0.00000
2027-04-05	1619.72265	1443.47407	-0.00008	0.00004	0.00000	4.92962	0.19116	-0.00032	-0.00002	0.00000
2027-04-13	1626.67199	1443.47574	0.00048	0.00004	0.00000	5.30838	0.18737	-0.00060	-0.00001	0.00000
2027-04-21	1633.62881	1443.48148	0.00096	0.00004	0.00000	5.67787	0.18199	-0.00077	-0.00002	0.00000
2027-04-29	1640.60116	1443.49131	0.00150	0.00005	0.00000	6.03494	0.17489	-0.00099	-0.00001	0.00000
2027-05-07	1647.59766	1443.50567	0.00208	0.00004	0.00000	6.37617	0.16616	-0.00121	-0.00002	0.00000
2027-05-15	1654.62719	1443.52426	0.00256	0.00004	0.00000	6.69826	0.15581	-0.00135	-0.00001	0.00000
2027-05-23	1661.69778	1443.54674	0.00306	0.00004	0.00000	6.99844	0.14420	-0.00156	-0.00002	0.00000
2027-05-31	1668.81753	1443.57348	0.00363	0.00005	0.00000	7.27389	0.13112	-0.00171	-0.00002	0.00000
2027-06-08	1675.99521	1443.60463	0.00413	0.00004	0.00000	7.52175	0.11658	-0.00189	0.00000	0.00000
2027-06-16	1683.23895	1443.63949	0.00459	0.00004	0.00000	7.73952	0.10109	-0.00201	-0.00002	0.00000
2027-06-24	1690.55613	1443.67809	0.00507	0.00004	0.00000	7.92510	0.08438	-0.00215	-0.00001	0.00000
2027-07-02	1697.95450	1443.72069	0.00556	0.00003	0.00000	8.07627	0.06666	-0.00230	-0.00001	0.00000
2027-07-10	1705.44143	1443.76652	0.00589	0.00003	0.00000	8.19106	0.04812	-0.00233	-0.00001	0.00000
2027-07-18	1713.02256	1443.81487	0.00620	0.00002	0.00000	8.26833	0.02907	-0.00243	0.00000	0.00000
2027-07-26	1720.70281	1443.86562	0.00648	0.00002	0.00000	8.30698	0.00957	-0.00245	-0.00001	0.00000
2027-08-03	1728.48639	1443.91804	0.00658	0.00000	0.00000	8.30636	-0.01018	-0.00244	0.00001	0.00000
2027-08-11	1736.37497	1443.97049	0.00652	-0.00001	0.00000	8.26676	-0.02938	-0.00238	0.00000	0.00000
2027-08-19	1744.36769	1444.02211	0.00637	-0.00002	0.00000	8.18927	-0.04802	-0.00226	0.00001	0.00000
2027-08-27	1752.46205	1444.07199	0.00605	-0.00004	0.00000	8.07559	-0.06556	-0.00212	0.00002	0.00000
2027-09-04	1760.65256	1444.11800	0.00543	-0.00006	0.00000	7.92850	-0.08127	-0.00181	0.00002	0.00000
2027-09-12	1768.92970	1444.15851	0.00468	-0.00007	0.00000	7.75231	-0.09469	-0.00152	0.00003	0.00000

Saturn 0h UT1	GHA c0	GHA c1	GHA c2	GHA c3	GHA c4	Dec c0	Dec c1	Dec c2	Dec c3	Dec c4
2027-09-20	1777.28148	1444.19254	0.00380	-0.00008	0.00000	7.55198	-0.10532	-0.00115	0.00003	0.00000
2027-09-28	1785.69350	1444.21852	0.00265	-0.00010	0.00000	7.33352	-0.11274	-0.00066	0.00005	0.00000
2027-10-06	1794.14777	1444.23470	0.00139	-0.00011	0.00000	7.10426	-0.11613	-0.00020	0.00004	0.00000
2027-10-14	1442.62430	1444.24079	0.00014	-0.00010	0.00000	6.87206	-0.11563	0.00033	0.00004	0.00000
2027-10-22	1451.10291	1444.23673	-0.00117	-0.00012	0.00000	6.64489	-0.11114	0.00081	0.00005	0.00000
2027-10-30	1459.56267	1444.22191	-0.00251	-0.00010	0.00000	6.43094	-0.10236	0.00134	0.00003	0.00000
2027-11-07	1467.98263	1444.19708	-0.00367	-0.00009	0.00000	6.23828	-0.08994	0.00177	0.00004	0.00000
2027-11-15	1476.34434	1444.16383	-0.00464	-0.00008	0.00000	6.07378	-0.07427	0.00211	0.00002	0.00000
2027-11-23	1484.63196	1444.12303	-0.00554	-0.00007	0.00000	5.94327	-0.05593	0.00247	0.00002	0.00000
2027-12-01	1492.83146	1444.07592	-0.00620	-0.00004	0.00000	5.85180	-0.03540	0.00265	0.00001	0.00000
2027-12-09	1500.93241	1444.02471	-0.00657	-0.00002	0.00000	5.80264	-0.01364	0.00278	0.00000	0.00000
2027-12-17	1508.92842	1443.97108	-0.00683	-0.00002	0.00000	5.79759	0.00861	0.00281	0.00001	0.00000
2027-12-25	1516.81546	1443.91589	-0.00693	0.00000	0.00000	5.83738	0.03116	0.00279	-0.00001	0.00000

Moon — daily blocks (span 24 h, n = 6; HP in arcminutes, 4 coefficients)

Moon 0h UT1		c0	c1	c2	c3	c4	c5	HP c0	HP c1	HP c2	HP c3
2026-09-01	GHA	492.19908	173.89939	-0.05595	-0.00097	0.00004	0.00000	58.252	0.203	-0.003	0.000
	Dec	16.60200	2.67210	-0.04802	-0.00191	-0.00001	0.00000				
2026-09-02	GHA	479.52197	173.41756	-0.06301	-0.00012	0.00007	0.00000	58.634	0.179	-0.003	0.000
	Dec	21.48783	2.19401	-0.07166	-0.00200	0.00000	0.00000				
2026-09-03	GHA	465.86364	172.92938	-0.05665	0.00124	0.00009	0.00000	58.965	0.152	-0.004	0.000
	Dec	25.22823	1.52719	-0.09464	-0.00176	0.00003	0.00000				
2026-09-04	GHA	451.33388	172.56049	-0.03314	0.00262	0.00007	0.00000	59.237	0.119	-0.005	0.000
	Dec	27.46569	0.69582	-0.11192	-0.00103	0.00006	0.00000				
2026-09-05	GHA	436.29884	172.43432	0.00251	0.00313	-0.00001	0.00000	59.433	0.076	-0.006	0.000
	Dec	27.93557	-0.23079	-0.11773	0.00010	0.00008	0.00000				
2026-09-06	GHA	421.30072	172.59608	0.03659	0.00237	-0.00008	0.00000	59.533	0.022	-0.008	0.000
	Dec	26.55035	-1.14751	-0.10946	0.00124	0.00006	0.00000				
2026-09-07	GHA	406.85905	172.97894	0.05658	0.00093	-0.00009	0.00000	59.512	-0.044	-0.009	0.000
	Dec	23.43687	-1.94952	-0.08977	0.00197	0.00003	0.00000				
2026-09-08	GHA	393.28822	173.45273	0.05963	-0.00035	-0.00006	0.00000	59.349	-0.119	-0.010	0.000
	Dec	18.89849	-2.56761	-0.06433	0.00221	0.00000	0.00000				
2026-09-09	GHA	380.64712	173.89852	0.05052	-0.00109	-0.00003	0.00000	59.035	-0.195	-0.009	0.000
	Dec	13.33217	-2.97687	-0.03814	0.00212	-0.00001	0.00000				
2026-09-10	GHA	368.80261	174.24439	0.03546	-0.00137	-0.00001	0.00000	58.574	-0.264	-0.008	0.000
	Dec	7.15122	-3.18400	-0.01406	0.00188	-0.00002	0.00000				
2026-09-11	GHA	357.52246	174.46159	0.01883	-0.00137	0.00001	0.00000	57.991	-0.317	-0.005	0.000
	Dec	0.73902	-3.21074	0.00690	0.00161	-0.00002	0.00000				
2026-09-12	GHA	346.54543	174.54825	0.00309	-0.00123	0.00001	0.00000	57.325	-0.346	-0.002	0.000
	Dec	-5.56916	-3.08259	0.02471	0.00136	-0.00001	0.00000				
2026-09-13	GHA	335.62246	174.51759	-0.01032	-0.00099	0.00002	0.00000	56.626	-0.350	0.001	0.000
	Dec	-11.48737	-2.82303	0.03983	0.00116	-0.00001	0.00000				
2026-09-14	GHA	324.54163	174.39341	-0.02010	-0.00063	0.00003	0.00000	55.947	-0.327	0.004	0.000
	Dec	-16.77269	-2.45152	0.05276	0.00100	-0.00001	0.00000				
2026-09-15	GHA	313.14937	174.21038	-0.02482	-0.00014	0.00003	0.00000	55.337	-0.281	0.007	0.000
	Dec	-21.21760	-1.98419	0.06379	0.00084	-0.00001	0.00000				
2026-09-16	GHA	301.37282	174.01441	-0.02319	0.00042	0.00004	0.00000	54.837	-0.217	0.009	0.000
	Dec	-24.64603	-1.43679	0.07272	0.00064	-0.00001	0.00000				
2026-09-17	GHA	289.23841	173.85800	-0.01502	0.00092	0.00003	0.00000	54.480	-0.139	0.010	0.000
	Dec	-26.91650	-0.82855	0.07888	0.00037	-0.00002	0.00000				
2026-09-18	GHA	276.87320	173.78757	-0.00213	0.00118	0.00001	0.00000	54.286	-0.055	0.011	0.000
	Dec	-27.93214	-0.18494	0.08145	0.00005	-0.00002	0.00000				
2026-09-19	GHA	264.47579	173.82658	0.01170	0.00107	-0.00002	0.00000	54.264	0.033	0.011	0.000
	Dec	-27.65259	0.46328	0.08002	-0.00028	-0.00002	0.00000				
2026-09-20	GHA	252.25893	173.96539	0.02227	0.00065	-0.00003	0.00000	54.414	0.117	0.010	0.000
	Dec	-26.10021	1.08481	0.07488	-0.00056	-0.00001	0.00000				
2026-09-21	GHA	240.38629	174.16581	0.02687	0.00011	-0.00003	0.00000	54.725	0.193	0.009	0.000
	Dec	-23.35546	1.65329	0.06690	-0.00076	-0.00001	0.00000				
2026-09-22	GHA	228.93051	174.37681	0.02498	-0.00041	-0.00003	0.00000	55.177	0.257	0.007	0.000
	Dec	-19.54410	2.14981	0.05700	-0.00089	-0.00001	0.00000				
2026-09-23	GHA	217.86332	174.54982	0.01758	-0.00081	-0.00002	0.00000	55.738	0.303	0.004	0.000
	Dec	-14.82363	2.56120	0.04565	-0.00100	-0.00001	0.00000				
2026-09-24	GHA	207.06920	174.64649	0.00610	-0.00109	-0.00001	0.00000	56.371	0.327	0.002	0.000
	Dec	-9.37578	2.87593	0.03279	-0.00115	-0.00001	0.00000				
2026-09-25	GHA	196.36704	174.63943	-0.00819	-0.00128	-0.00001	0.00000	57.028	0.328	-0.001	0.000
	Dec	-3.40726	3.08023	0.01795	-0.00134	-0.00001	0.00000				
2026-09-26	GHA	185.53034	174.51051	-0.02420	-0.00137	0.00000	0.00000	57.664	0.305	-0.004	0.000

Moon 0h UT1		c0	c1	c2	c3	c4	c5	HP c0	HP c1	HP c2	HP c3
	Dec	2.84322	3.15571	0.00049	-0.00158	-0.00002	0.00000				
2026-09-27	GHA	174.30572	174.25126	-0.04050	-0.00131	0.00001	0.00000	58.233	0.262	-0.006	0.000
	Dec	9.09536	3.07929	-0.02007	-0.00185	-0.00002	0.00000				
2026-09-28	GHA	522.43768	173.86907	-0.05441	-0.00095	0.00004	0.00000	58.702	0.205	-0.008	0.000
	Dec	15.02012	2.82576	-0.04371	-0.00208	-0.00001	0.00000				
2026-09-29	GHA	509.71310	173.40062	-0.06122	-0.00010	0.00007	0.00000	59.048	0.141	-0.008	0.000
	Dec	20.24159	2.37442	-0.06922	-0.00213	0.00001	0.00000				
2026-09-30	GHA	496.03563	172.92725	-0.05477	0.00123	0.00009	0.00000	59.266	0.077	-0.008	0.000
	Dec	24.35791	1.72168	-0.09343	-0.00183	0.00003	0.00000				
2026-10-01	GHA	481.51606	172.57262	-0.03150	0.00258	0.00007	0.00000	59.362	0.020	-0.007	0.000
	Dec	26.99243	0.89810	-0.11107	-0.00103	0.00006	0.00000				
2026-10-02	GHA	466.51689	172.45763	0.00364	0.00308	-0.00001	0.00000	59.351	-0.029	-0.006	0.000
	Dec	27.87401	-0.02127	-0.11676	0.00011	0.00007	0.00000				
2026-10-03	GHA	451.57269	172.62625	0.03720	0.00234	-0.00008	0.00000	59.250	-0.071	-0.005	0.000
	Dec	26.91534	-0.93065	-0.10870	0.00118	0.00006	0.00000				
2026-10-04	GHA	437.19526	173.01278	0.05693	0.00092	-0.00009	0.00000	59.072	-0.107	-0.004	0.000
	Dec	24.23858	-1.73062	-0.09017	0.00182	0.00002	0.00000				
2026-10-05	GHA	423.69482	173.48928	0.05998	-0.00034	-0.00006	0.00000	58.826	-0.140	-0.004	0.000
	Dec	20.12856	-2.35993	-0.06685	0.00200	0.00000	0.00000				
2026-10-06	GHA	411.12994	173.93828	0.05097	-0.00108	-0.00003	0.00000	58.513	-0.173	-0.004	0.000
	Dec	14.94941	-2.79951	-0.04322	0.00191	-0.00001	0.00000				
2026-10-07	GHA	399.36880	174.28803	0.03596	-0.00137	-0.00001	0.00000	58.134	-0.206	-0.004	0.000
	Dec	9.07510	-3.05662	-0.02139	0.00172	-0.00001	0.00000				
2026-10-08	GHA	388.17976	174.50898	0.01924	-0.00139	0.00000	0.00000	57.692	-0.236	-0.004	0.000
	Dec	2.85410	-3.14805	-0.00178	0.00155	-0.00001	0.00000				
2026-10-09	GHA	377.30012	174.59803	0.00324	-0.00126	0.00001	0.00000	57.193	-0.261	-0.003	0.000
	Dec	-3.39935	-3.09069	0.01587	0.00140	-0.00001	0.00000				
2026-10-10	GHA	366.47666	174.56696	-0.01060	-0.00103	0.00002	0.00000	56.654	-0.276	-0.001	0.000
	Dec	-9.40204	-2.89860	0.03194	0.00128	-0.00001	0.00000				
2026-10-11	GHA	355.49049	174.43822	-0.02097	-0.00068	0.00003	0.00000	56.099	-0.277	0.001	0.000
	Dec	-14.89645	-2.58358	0.04660	0.00116	-0.00001	0.00000				
2026-10-12	GHA	344.17886	174.24569	-0.02631	-0.00019	0.00003	0.00000	55.558	-0.262	0.003	0.000
	Dec	-19.64835	-2.15743	0.05967	0.00101	-0.00001	0.00000				
2026-10-13	GHA	332.45939	174.03582	-0.02513	0.00040	0.00004	0.00000	55.066	-0.228	0.005	0.000
	Dec	-23.45003	-1.63522	0.07049	0.00078	-0.00002	0.00000				
2026-10-14	GHA	320.35198	173.86359	-0.01696	0.00094	0.00003	0.00000	54.658	-0.178	0.007	0.000
	Dec	-26.13026	-1.03868	0.07809	0.00047	-0.00002	0.00000				
2026-10-15	GHA	307.98394	173.77963	-0.00349	0.00125	0.00001	0.00000	54.364	-0.114	0.009	0.000
	Dec	-27.56954	-0.39777	0.08148	0.00009	-0.00002	0.00000				
2026-10-16	GHA	295.56279	173.81162	0.01134	0.00116	-0.00002	0.00000	54.212	-0.038	0.010	0.000
	Dec	-27.71441	0.25192	0.08030	-0.00028	-0.00002	0.00000				
2026-10-17	GHA	283.31663	173.95194	0.02298	0.00074	-0.00003	0.00000	54.220	0.046	0.011	0.000
	Dec	-26.58244	0.87577	0.07517	-0.00056	-0.00001	0.00000				
2026-10-18	GHA	271.42555	174.16148	0.02838	0.00016	-0.00004	0.00000	54.400	0.133	0.011	0.000
	Dec	-24.25303	1.44714	0.06739	-0.00072	-0.00001	0.00000				
2026-10-19	GHA	259.97469	174.38641	0.02689	-0.00039	-0.00003	0.00000	54.752	0.219	0.010	0.000
	Dec	-20.84814	1.95011	0.05823	-0.00080	0.00000	0.00000				
2026-10-20	GHA	248.94240	174.57516	0.01955	-0.00081	-0.00002	0.00000	55.268	0.296	0.009	0.000
	Dec	-16.51287	2.37697	0.04840	-0.00085	0.00000	0.00000				
2026-10-21	GHA	238.21432	174.68702	0.00788	-0.00112	-0.00002	0.00000	55.924	0.358	0.007	0.000
	Dec	-11.40477	2.72229	0.03777	-0.00094	-0.00001	0.00000				
2026-10-22	GHA	227.60621	174.69258	-0.00686	-0.00133	-0.00001	0.00000	56.684	0.399	0.003	0.000
	Dec	-5.69543	2.97689	0.02556	-0.00112	-0.00001	0.00000				

Moon 0h UT1		c0	c1	c2	c3	c4	c5	HP c0	HP c1	HP c2	HP c3
2026-10-23	GHA	216.88401	174.57116	-0.02373	-0.00147	-0.00001	0.00000	57.497	0.410	-0.001	0.000
	Dec	0.41724	3.12325	0.01052	-0.00141	-0.00002	0.00000				
2026-10-24	GHA	205.78016	174.31011	-0.04155	-0.00147	0.00001	0.00000	58.299	0.388	-0.005	0.000
	Dec	6.68994	3.13349	-0.00864	-0.00180	-0.00003	0.00000				
2026-10-25	GHA	194.01428	173.91033	-0.05792	-0.00119	0.00003	0.00000	59.023	0.332	-0.009	0.000
	Dec	12.81452	2.97104	-0.03269	-0.00220	-0.00002	0.00000				
2026-10-26	GHA	181.33412	173.40105	-0.06799	-0.00039	0.00007	0.00000	59.602	0.245	-0.012	0.000
	Dec	18.40756	2.59845	-0.06091	-0.00246	-0.00001	0.00000				
2026-10-27	GHA	527.59347	172.86124	-0.06439	0.00108	0.00011	0.00000	59.988	0.139	-0.014	0.000
	Dec	23.02428	1.99406	-0.08989	-0.00228	0.00003	0.00000				
2026-10-28	GHA	512.86257	172.42728	-0.04111	0.00277	0.00009	0.00000	60.156	0.028	-0.014	0.000
	Dec	26.21450	1.17687	-0.11295	-0.00145	0.00007	0.00000				
2026-10-29	GHA	497.50734	172.25058	-0.00158	0.00360	0.00000	-0.00001	60.109	-0.074	-0.012	0.000
	Dec	27.62489	0.22556	-0.12247	-0.00008	0.00009	0.00000				
2026-10-30	GHA	482.12859	172.40444	0.03877	0.00289	-0.00009	0.00000	59.875	-0.157	-0.009	0.000
	Dec	27.11011	-0.73418	-0.11512	0.00125	0.00007	0.00000				
2026-10-31	GHA	467.33853	172.82636	0.06376	0.00122	-0.00011	0.00000	59.500	-0.216	-0.006	0.000
	Dec	24.77960	-1.57937	-0.09484	0.00202	0.00003	0.00000				
2026-11-01	GHA	453.52784	173.36703	0.06873	-0.00030	-0.00008	0.00000	59.031	-0.251	-0.003	0.000
	Dec	20.94199	-2.23692	-0.06932	0.00215	-0.00001	0.00000				
2026-11-02	GHA	440.78774	173.88485	0.05919	-0.00118	-0.00004	0.00000	58.511	-0.267	-0.001	0.000
	Dec	15.99347	-2.69097	-0.04458	0.00194	-0.00002	0.00000				
2026-11-03	GHA	428.98087	174.29444	0.04264	-0.00151	-0.00001	0.00000	57.972	-0.270	0.000	0.000
	Dec	10.32492	-2.95970	-0.02312	0.00164	-0.00002	0.00000				
2026-11-04	GHA	417.85274	174.56207	0.02424	-0.00153	0.00000	0.00000	57.435	-0.266	0.001	0.000
	Dec	4.27990	-3.07015	-0.00491	0.00141	-0.00001	0.00000				
2026-11-05	GHA	407.11409	174.68459	0.00662	-0.00139	0.00001	0.00000	56.911	-0.258	0.001	0.000
	Dec	-1.84813	-3.04459	0.01105	0.00127	-0.00001	0.00000				
2026-11-06	GHA	396.48567	174.67408	-0.00886	-0.00117	0.00002	0.00000	56.406	-0.248	0.001	0.000
	Dec	-7.80171	-2.89671	0.02580	0.00120	0.00000	0.00000				
2026-11-07	GHA	385.72212	174.55220	-0.02103	-0.00084	0.00003	0.00000	55.922	-0.236	0.002	0.000
	Dec	-13.34358	-2.63334	0.03998	0.00116	0.00000	0.00000				
2026-11-08	GHA	374.63171	174.35126	-0.02837	-0.00036	0.00003	0.00000	55.466	-0.220	0.002	0.000
	Dec	-18.24690	-2.25864	0.05358	0.00110	-0.00001	0.00000				
2026-11-09	GHA	363.10077	174.11721	-0.02906	0.00026	0.00004	0.00000	55.045	-0.199	0.003	0.000
	Dec	-22.29542	-1.77962	0.06589	0.00094	-0.00001	0.00000				
2026-11-10	GHA	351.12057	173.90851	-0.02198	0.00091	0.00004	0.00000	54.675	-0.170	0.004	0.000
	Dec	-25.29504	-1.21197	0.07550	0.00064	-0.00002	0.00000				
2026-11-11	GHA	338.80244	173.78493	-0.00814	0.00135	0.00002	0.00000	54.374	-0.130	0.006	0.000
	Dec	-27.09519	-0.58373	0.08083	0.00023	-0.00003	0.00000				
2026-11-12	GHA	326.36009	173.78675	0.00864	0.00138	-0.00001	0.00000	54.164	-0.079	0.007	0.000
	Dec	-27.61265	0.06633	0.08091	-0.00021	-0.00003	0.00000				
2026-11-13	GHA	314.05124	173.91647	0.02308	0.00098	-0.00003	0.00000	54.068	-0.017	0.008	0.000
	Dec	-26.84550	0.69677	0.07607	-0.00057	-0.00002	0.00000				
2026-11-14	GHA	302.09871	174.13772	0.03112	0.00035	0.00004	0.00000	54.105	0.055	0.010	0.000
	Dec	-24.86800	1.27386	0.06785	-0.00077	-0.00001	0.00000				
2026-11-15	GHA	290.62867	174.39253	0.03149	-0.00027	-0.00004	0.00000	54.295	0.135	0.010	0.000
	Dec	-21.80798	1.77801	0.05809	-0.00083	0.00000	0.00000				
2026-11-16	GHA	279.64901	174.62251	0.02515	-0.00076	-0.00003	0.00000	54.649	0.219	0.011	0.000
	Dec	-17.81889	2.20277	0.04813	-0.00082	0.00000	0.00000				
2026-11-17	GHA	269.06191	174.78097	0.01386	-0.00110	-0.00002	0.00000	55.170	0.302	0.010	0.000
	Dec	-13.05930	2.54868	0.03836	-0.00081	0.00000	0.00000				
2026-11-18	GHA	258.68970	174.83449	-0.00092	-0.00135	-0.00001	0.00000	55.850	0.377	0.009	0.000

Moon 0h UT1		c0	c1	c2	c3	c4	c5	HP c0	HP c1	HP c2	HP c3
	Dec	-7.68648	2.81575	0.02828	-0.00088	-0.00001	0.00000				
2026-11-19	GHA	248.29766	174.75899	-0.01829	-0.00154	-0.00001	0.00000	56.665	0.436	0.006	0.000
	Dec	-1.86425	2.99680	0.01664	-0.00108	-0.00002	0.00000				
2026-11-20	GHA	237.60916	174.53646	-0.03755	-0.00165	0.00000	0.00000	57.574	0.470	0.002	0.000
	Dec	4.21759	3.07260	0.00169	-0.00144	-0.00003	0.00000				
2026-11-21	GHA	226.31889	174.15683	-0.05724	-0.00158	0.00002	0.00000	58.515	0.468	-0.003	0.000
	Dec	10.31610	3.00916	-0.01842	-0.00193	-0.00003	0.00000				
2026-11-22	GHA	214.11911	173.62965	-0.07365	-0.00106	0.00005	0.00000	59.411	0.423	-0.008	0.000
	Dec	16.10718	2.75992	-0.04481	-0.00245	-0.00003	0.00000				
2026-11-23	GHA	200.76219	173.00859	-0.07934	0.00025	0.00011	0.00000	60.171	0.334	-0.014	0.000
	Dec	21.17097	2.27765	-0.07622	-0.00271	0.00000	0.00000				
2026-11-24	GHA	186.17703	172.41849	-0.06461	0.00228	0.00013	0.00000	60.715	0.207	-0.018	0.000
	Dec	25.01634	1.54204	-0.10693	-0.00227	0.00006	0.00000				
2026-11-25	GHA	530.60747	172.04436	-0.02585	0.00401	0.00007	-0.00001	60.981	0.058	-0.019	0.000
	Dec	27.17235	0.59724	-0.12691	-0.00093	0.00011	0.00000				
2026-11-26	GHA	514.64806	172.03826	0.02434	0.00402	-0.00006	-0.00001	60.947	-0.091	-0.018	0.000
	Dec	27.33739	-0.43262	-0.12743	0.00084	0.00010	0.00000				
2026-11-27	GHA	499.05465	172.40087	0.06325	0.00229	-0.00013	0.00000	60.633	-0.220	-0.014	0.000
	Dec	25.50275	-1.38633	-0.10871	0.00215	0.00005	0.00000				
2026-11-28	GHA	484.42373	172.98049	0.07805	0.00024	-0.00011	0.00000	60.095	-0.315	-0.009	0.000
	Dec	21.94947	-2.14267	-0.07967	0.00256	0.00000	0.00000				
2026-11-29	GHA	470.99986	173.59065	0.07217	-0.00108	-0.00006	0.00000	59.405	-0.371	-0.005	0.000
	Dec	17.12241	-2.65962	-0.04999	0.00232	-0.00003	0.00000				
2026-11-30	GHA	458.70911	174.10453	0.05536	-0.00162	-0.00002	0.00000	58.640	-0.391	-0.001	0.000
	Dec	11.48617	-2.95563	-0.02481	0.00187	-0.00003	0.00000				
2026-12-01	GHA	447.29768	174.46718	0.03517	-0.00170	0.00000	0.00000	57.865	-0.382	0.002	0.000
	Dec	5.44227	-3.07171	-0.00493	0.00146	-0.00002	0.00000				
2026-12-02	GHA	436.45004	174.66874	0.01545	-0.00157	0.00001	0.00000	57.127	-0.355	0.004	0.000
	Dec	-0.68858	-3.04597	0.01090	0.00120	-0.00001	0.00000				
2026-12-03	GHA	425.85381	174.72036	-0.00215	-0.00135	0.00002	0.00000	56.455	-0.317	0.005	0.000
	Dec	-6.64967	-2.90389	0.02440	0.00107	0.00000	0.00000				
2026-12-04	GHA	415.22935	174.64310	-0.01663	-0.00104	0.00002	0.00000	55.863	-0.275	0.005	0.000
	Dec	-12.22224	-2.65823	0.03696	0.00103	0.00000	0.00000				
2026-12-05	GHA	404.34763	174.46679	-0.02669	-0.00061	0.00003	0.00000	55.356	-0.233	0.005	0.000
	Dec	-17.20387	-2.31312	0.04930	0.00102	0.00000	0.00000				
2026-12-06	GHA	393.05132	174.23367	-0.03055	-0.00001	0.00004	0.00000	54.929	-0.194	0.005	0.000
	Dec	-21.39735	-1.87038	0.06127	0.00096	-0.00001	0.00000				
2026-12-07	GHA	381.28212	174.00047	-0.02653	0.00068	0.00004	0.00000	54.580	-0.156	0.005	0.000
	Dec	-24.61352	-1.33707	0.07172	0.00076	-0.00002	0.00000				
2026-12-08	GHA	369.10455	173.83196	-0.01456	0.00127	0.00003	0.00000	54.305	-0.119	0.005	0.000
	Dec	-26.68885	-0.73230	0.07885	0.00040	-0.00003	0.00000				
2026-12-09	GHA	356.70447	173.78225	0.00253	0.00150	0.00000	0.00000	54.105	-0.080	0.005	0.000
	Dec	-27.51266	-0.08972	0.08099	-0.00006	-0.00003	0.00000				
2026-12-10	GHA	344.34453	173.87195	0.01945	0.00125	-0.00003	0.00000	53.986	-0.038	0.006	0.000
	Dec	-27.05185	0.54769	0.07760	-0.00049	-0.00002	0.00000				
2026-12-11	GHA	332.28488	174.07809	0.03104	0.00065	-0.00004	0.00000	53.959	0.011	0.007	0.000
	Dec	-25.35855	1.13901	0.06971	-0.00079	-0.00001	0.00000				
2026-12-12	GHA	320.70601	174.34616	0.03479	-0.00002	-0.00004	0.00000	54.037	0.067	0.008	0.000
	Dec	-22.55498	1.65586	0.05929	-0.00092	0.00000	0.00000				
2026-12-13	GHA	309.66870	174.61338	0.03103	-0.00058	-0.00003	0.00000	54.234	0.131	0.008	0.000
	Dec	-18.80405	2.08584	0.04822	-0.00091	0.00000	0.00000				
2026-12-14	GHA	299.11639	174.82638	0.02152	-0.00098	-0.00002	0.00000	54.567	0.202	0.009	0.000
	Dec	-14.28067	2.42868	0.03758	-0.00086	0.00000	0.00000				

Moon 0h UT1		c0	c1	c2	c3	c4	c5	HP c0	HP c1	HP c2	HP c3
2026-12-15	GHA	288.90053	174.94651	0.00803	-0.00126	-0.00001	0.00000	55.046	0.277	0.009	0.000
	Dec	-9.15477	2.68882	0.02750	-0.00083	0.00000	0.00000				
2026-12-16	GHA	278.80738	174.94669	-0.00835	-0.00147	-0.00001	0.00000	55.675	0.351	0.009	0.000
	Dec	-3.58914	2.86828	0.01726	-0.00089	-0.00001	0.00000				
2026-12-17	GHA	268.57601	174.80636	-0.02704	-0.00164	-0.00001	0.00000	56.445	0.418	0.008	0.000
	Dec	2.24959	2.96065	0.00548	-0.00109	-0.00002	0.00000				
2026-12-18	GHA	257.90866	174.50931	-0.04738	-0.00173	0.00000	0.00000	57.335	0.469	0.005	0.000
	Dec	8.16930	2.94645	-0.00968	-0.00146	-0.00003	0.00000				
2026-12-19	GHA	246.48348	174.04872	-0.06747	-0.00156	-0.00002	0.00000	58.298	0.492	0.001	0.000
	Dec	13.92335	2.79050	-0.03022	-0.00198	-0.00004	0.00000				
2026-12-20	GHA	233.98884	173.44418	-0.08234	-0.00080	0.00007	0.00000	59.270	0.476	-0.005	-0.001
	Dec	19.18082	2.44445	-0.05722	-0.00250	-0.00003	0.00000				
2026-12-21	GHA	220.20550	172.77202	-0.08278	0.00087	0.00013	0.00000	60.164	0.412	-0.011	-0.001
	Dec	23.51323	1.86157	-0.08881	-0.00267	0.00001	0.00000				
2026-12-22	GHA	205.14711	172.18950	-0.05889	0.00312	0.00013	0.00000	60.881	0.300	-0.017	0.000
	Dec	26.42970	1.03068	-0.11774	-0.00199	0.00008	0.00000				
2026-12-23	GHA	189.19490	171.89791	-0.01156	0.00450	0.00002	-0.00001	61.333	0.148	-0.021	0.000
	Dec	27.49083	0.01804	-0.13255	-0.00036	-0.00012	0.00000				
2026-12-24	GHA	173.06656	172.01692	0.03999	0.00375	-0.00011	0.00000	61.457	-0.024	-0.022	0.000
	Dec	26.47599	-1.02689	-0.12539	0.00150	0.00010	0.00000				
2026-12-25	GHA	517.53875	172.48248	0.07257	0.00159	-0.00014	0.00000	61.240	-0.191	-0.019	0.000
	Dec	23.49287	-1.93458	-0.09952	0.00265	0.00004	0.00000				
2026-12-26	GHA	503.11901	173.10319	0.07917	-0.00037	-0.00009	0.00000	60.716	-0.328	-0.015	0.000
	Dec	18.93272	-2.59748	-0.06591	0.00282	-0.00001	0.00000				
2026-12-27	GHA	489.92958	173.69767	0.06763	-0.00141	-0.00004	0.00000	59.961	-0.421	-0.009	0.001
	Dec	13.31381	-2.99496	-0.03419	0.00241	-0.00003	0.00000				
2026-12-28	GHA	477.80683	174.16336	0.04824	-0.00174	-0.00001	0.00000	59.069	-0.466	-0.003	0.000
	Dec	7.13536	-3.16215	-0.00858	0.00186	-0.00003	0.00000				
2026-12-29	GHA	466.45335	174.46584	0.02747	-0.00169	0.00001	0.00000	58.131	-0.468	0.002	0.000
	Dec	0.80701	-3.15005	0.01080	0.00140	-0.00002	0.00000				
2026-12-30	GHA	455.54261	174.60742	0.00828	-0.00149	0.00002	0.00000	57.221	-0.438	0.005	0.000
	Dec	-5.35757	-3.00216	0.02566	0.00110	-0.00001	0.00000				
2026-12-31	GHA	444.77015	174.60655	-0.00800	-0.00121	0.00002	0.00000	56.394	-0.387	0.007	0.000
	Dec	-11.11683	-2.74691	0.03790	0.00096	-0.00001	0.00000				
2027-01-01	GHA	433.87776	174.49084	-0.02025	-0.00082	0.00003	0.00000	55.681	-0.325	0.008	0.000
	Dec	-16.27186	-2.39887	0.04901	0.00090	0.00000	0.00000				
2027-01-02	GHA	422.67242	174.29819	-0.02698	-0.00028	0.00004	0.00000	55.096	-0.260	0.008	0.000
	Dec	-20.64375	-1.96420	0.05959	0.00086	0.00000	0.00000				
2027-01-03	GHA	411.04981	174.07958	-0.02652	0.00037	0.00004	0.00000	54.641	-0.196	0.008	0.000
	Dec	-24.06401	-1.44799	0.06926	0.00074	-0.00001	0.00000				
2027-01-04	GHA	399.01899	173.89663	-0.01810	0.00101	0.00003	0.00000	54.308	-0.137	0.007	0.000
	Dec	-26.38055	-0.86231	0.07671	0.00048	-0.00002	0.00000				
2027-01-05	GHA	386.71157	173.80831	-0.00331	0.00139	0.00001	0.00000	54.086	-0.085	0.006	0.000
	Dec	-27.47756	-0.23174	0.08025	0.00009	-0.00003	0.00000				
2027-01-06	GHA	374.35538	173.84958	0.01353	0.00134	-0.00002	0.00000	53.965	-0.037	0.006	0.000
	Dec	-27.30084	0.40717	0.07871	-0.00034	-0.00003	0.00000				
2027-01-07	GHA	362.20915	174.01559	0.02717	0.00088	-0.00004	0.00000	53.934	0.006	0.005	0.000
	Dec	-25.87459	1.01370	0.07228	-0.00071	-0.00002	0.00000				
2027-01-08	GHA	350.48386	174.26485	0.03403	0.00025	-0.00004	0.00000	53.989	0.048	0.005	0.000
	Dec	-23.29891	1.55370	0.06235	-0.00092	-0.00001	0.00000				
2027-01-09	GHA	339.28796	174.53872	0.03340	-0.00034	-0.00003	0.00000	54.128	0.091	0.006	0.000
	Dec	-19.72897	2.00657	0.05076	-0.00099	0.00000	0.00000				
2027-01-10	GHA	328.61401	174.78161	0.02655	-0.00078	-0.00002	0.00000	54.356	0.137	0.006	0.000

Moon 0h UT1		c0	c1	c2	c3	c4	c5	HP c0	HP c1	HP c2	HP c3
	Dec	-15.34731	2.36525	0.03896	-0.00097	0.00000	0.00000				
2027-01-11	GHA	318.35601	174.95098	0.01526	-0.00108	-0.00002	0.00000	54.681	0.188	0.007	0.000
	Dec	-10.34136	2.63126	0.02763	-0.00092	0.00000	0.00000				
2027-01-12	GHA	308.33615	175.01720	0.00091	-0.00130	-0.00001	0.00000	55.111	0.243	0.007	0.000
	Dec	-4.89262	2.80836	0.01667	-0.00091	0.00000	0.00000				
2027-01-13	GHA	298.32634	174.95910	-0.01574	-0.00147	-0.00001	0.00000	55.655	0.301	0.007	0.000
	Dec	0.82206	2.89688	0.00533	-0.00099	-0.00001	0.00000				
2027-01-14	GHA	288.06102	174.76018	-0.03421	-0.00160	-0.00001	0.00000	56.314	0.358	0.007	0.000
	Dec	6.61873	2.88900	-0.00765	-0.00119	-0.00002	0.00000				
2027-01-15	GHA	277.24652	174.40921	-0.05352	-0.00159	0.00001	0.00000	57.080	0.407	0.005	0.000
	Dec	12.28651	2.76526	-0.02389	-0.00153	-0.00003	0.00000				
2027-01-16	GHA	265.57928	173.90900	-0.07092	-0.00124	0.00004	0.00000	57.930	0.441	0.003	0.000
	Dec	17.56247	2.49320	-0.04490	-0.00197	-0.00003	0.00000				
2027-01-17	GHA	252.79275	173.29646	-0.08045	-0.00023	0.00009	0.00000	58.823	0.448	-0.001	0.000
	Dec	22.10967	2.03225	-0.07097	-0.00234	-0.00001	0.00000				
2027-01-18	GHA	238.75289	172.66974	-0.07310	0.00155	0.00013	0.00000	59.692	0.418	-0.006	0.000
	Dec	25.51676	1.35131	-0.09913	-0.00225	0.00003	0.00000				
2027-01-19	GHA	223.59050	172.19328	-0.04268	0.00343	0.00009	0.00000	60.458	0.343	-0.012	0.000
	Dec	27.34890	0.46198	-0.12164	-0.00136	0.00008	0.00000				
2027-01-20	GHA	207.77877	172.03404	0.00395	0.00405	-0.00002	-0.00001	61.028	0.223	-0.017	0.000
	Dec	27.26616	-0.55062	-0.12867	0.00025	0.00011	0.00000				
2027-01-21	GHA	192.02207	172.24497	0.04662	0.00283	-0.00012	0.00000	61.323	0.069	-0.021	0.000
	Dec	25.16551	-1.53915	-0.11566	0.00185	0.00008	0.00000				
2027-01-22	GHA	176.96862	172.71990	0.06860	0.00083	-0.00012	0.00000	61.293	-0.099	-0.021	0.000
	Dec	21.24509	-2.35766	-0.08744	0.00272	0.00003	0.00000				
2027-01-23	GHA	522.96822	173.27967	0.06866	-0.00070	-0.00007	0.00000	60.934	-0.257	-0.018	0.000
	Dec	15.93653	-2.92272	-0.05370	0.00280	0.00001	0.00000				
2027-01-24	GHA	510.03925	173.78003	0.05515	-0.00144	-0.00003	0.00000	60.291	-0.382	-0.013	0.000
	Dec	9.76397	-3.22357	-0.02220	0.00241	-0.00003	0.00000				
2027-01-25	GHA	497.98250	174.14757	0.03643	-0.00162	0.00000	0.00000	59.444	-0.460	-0.007	0.001
	Dec	3.22450	-3.29442	0.00358	0.00189	-0.00003	0.00000				
2027-01-26	GHA	486.50824	174.36234	0.01745	-0.00151	0.00001	0.00000	58.487	-0.491	-0.001	0.000
	Dec	-3.26977	-3.18333	0.02340	0.00144	-0.00002	0.00000				
2027-01-27	GHA	475.31766	174.43308	0.00066	-0.00127	0.00002	0.00000	57.514	-0.479	0.004	0.000
	Dec	-9.39889	-2.93316	0.03858	0.00112	-0.00002	0.00000				
2027-01-28	GHA	464.14493	174.38319	-0.01251	-0.00091	0.00003	0.00000	56.598	-0.435	0.007	0.000
	Dec	-14.91672	-2.57457	0.05072	0.00092	-0.00001	0.00000				
2027-01-29	GHA	452.78198	174.24703	-0.02069	-0.00043	0.00003	0.00000	55.791	-0.370	0.009	0.000
	Dec	-19.62666	-2.12681	0.06099	0.00079	-0.00001	0.00000				
2027-01-30	GHA	441.10090	174.07044	-0.02242	0.00016	0.00004	0.00000	55.127	-0.294	0.010	0.000
	Dec	-23.36366	-1.60289	0.06975	0.00066	-0.00001	0.00000				
2027-01-31	GHA	429.07574	173.90903	-0.01688	0.00076	0.00003	0.00000	54.619	-0.214	0.010	0.000
	Dec	-25.98874	-1.01660	0.07645	0.00044	-0.00002	0.00000				
2027-02-01	GHA	416.79333	173.81847	-0.00502	0.00117	0.00002	0.00000	54.267	-0.138	0.009	0.000
	Dec	-27.39697	-0.38869	0.07998	0.00013	-0.00002	0.00000				
2027-02-02	GHA	404.43662	173.83723	0.00982	0.00124	-0.00001	0.00000	54.064	-0.067	0.008	0.000
	Dec	-27.53401	0.25111	0.07932	-0.00024	-0.00002	0.00000				
2027-02-03	GHA	392.23376	173.97099	0.02306	0.00092	-0.00003	0.00000	53.993	-0.004	0.007	0.000
	Dec	-26.41073	0.86798	0.07429	-0.00059	-0.00002	0.00000				
2027-02-04	GHA	380.38916	174.19104	0.03101	0.00039	-0.00004	0.00000	54.039	0.049	0.006	0.000
	Dec	-24.10593	1.42964	0.06569	-0.00083	-0.00001	0.00000				
2027-02-05	GHA	369.02716	174.44800	0.03227	-0.00017	-0.00003	0.00000	54.183	0.094	0.005	0.000
	Dec	-20.75457	1.91268	0.05484	-0.00096	0.00000	0.00000				

Moon 0h UT1		c0	c1	c2	c3	c4	c5	HP c0	HP c1	HP c2	HP c3
2027-02-06	GHA	358.16933	174.69018	0.02750	-0.00061	-0.00002	0.00000	54.409	0.132	0.004	0.000
	Dec	-16.52767	2.30434	0.04300	-0.00100	0.00000	0.00000				
2027-02-07	GHA	347.74252	174.87527	0.01823	-0.00092	-0.00002	0.00000	54.706	0.165	0.004	0.000
	Dec	-11.61327	2.59999	0.03091	-0.00101	0.00000	0.00000				
2027-02-08	GHA	337.60112	174.97300	0.00583	-0.00113	-0.00001	0.00000	55.067	0.195	0.004	0.000
	Dec	-6.20432	2.79886	0.01879	-0.00102	0.00000	0.00000				
2027-02-09	GHA	327.54873	174.96246	-0.00874	-0.00129	-0.00001	0.00000	55.488	0.226	0.004	0.000
	Dec	-0.49528	2.89983	0.00637	-0.00106	0.00000	0.00000				
2027-02-10	GHA	317.35335	174.82870	-0.02488	-0.00139	0.00000	0.00000	55.970	0.257	0.004	0.000
	Dec	5.31367	2.89794	-0.00705	-0.00118	-0.00001	0.00000				
2027-02-11	GHA	306.75828	174.56215	-0.04177	-0.00140	0.00000	0.00000	56.514	0.288	0.004	0.000
	Dec	11.00600	2.78160	-0.02240	-0.00139	-0.00002	0.00000				
2027-02-12	GHA	295.49704	174.16345	-0.05752	-0.00118	0.00003	0.00000	57.121	0.318	0.003	0.000
	Dec	16.33400	2.53115	-0.04070	-0.00167	-0.00002	0.00000				
2027-02-13	GHA	283.32545	173.65600	-0.06819	-0.00052	0.00006	0.00000	57.780	0.340	0.002	0.000
	Dec	21.00402	2.12081	-0.06234	-0.00192	-0.00001	0.00000				
2027-02-14	GHA	270.08553	173.10461	-0.06749	0.00072	0.00009	0.00000	58.471	0.349	0.000	0.000
	Dec	24.67236	1.52780	-0.08597	-0.00196	0.00001	0.00000				
2027-02-15	GHA	255.80080	172.62561	-0.04954	0.00226	0.00009	0.00000	59.159	0.336	-0.003	0.000
	Dec	26.96978	0.75187	-0.10720	-0.00149	0.00005	0.00000				
2027-02-16	GHA	240.75637	172.35843	-0.01552	0.00324	0.00002	0.00000	59.792	0.293	-0.007	0.000
	Dec	27.57064	-0.16104	-0.11916	-0.00043	0.00008	0.00000				
2027-02-17	GHA	225.47201	172.38909	0.02256	0.00289	-0.00006	0.00000	60.305	0.216	-0.012	0.000
	Dec	26.29494	-1.11227	-0.11630	0.00090	0.00008	0.00000				
2027-02-18	GHA	210.52532	172.68659	0.04934	0.00149	-0.00010	0.00000	60.631	0.107	-0.015	0.000
	Dec	23.18854	-1.97931	-0.09858	0.00197	0.00005	0.00000				
2027-02-19	GHA	196.33088	173.12610	0.05792	0.00000	-0.00008	0.00000	60.714	-0.025	-0.017	0.000
	Dec	18.52408	-2.66234	-0.07127	0.00249	0.00001	0.00000				
2027-02-20	GHA	183.03311	173.57077	0.05156	-0.00096	-0.00004	0.00000	60.525	-0.164	-0.017	0.000
	Dec	12.72518	-3.11122	-0.04091	0.00251	-0.00001	0.00000				
2027-02-21	GHA	530.54429	173.92833	0.03713	-0.00137	-0.00001	0.00000	60.070	-0.289	-0.014	0.000
	Dec	6.26818	-3.32185	-0.01222	0.00224	-0.00002	0.00000				
2027-02-22	GHA	518.64468	174.15795	0.02022	-0.00141	0.00001	0.00000	59.393	-0.384	-0.010	0.000
	Dec	-0.39238	-3.31811	0.01250	0.00187	-0.00002	0.00000				
2027-02-23	GHA	507.07045	174.25449	0.00423	-0.00123	0.00002	0.00000	58.565	-0.439	-0.004	0.000
	Dec	-6.86211	-3.13480	0.03269	0.00150	-0.00002	0.00000				
2027-02-24	GHA	495.56986	174.23408	-0.00876	-0.00091	0.00002	0.00000	57.668	-0.453	0.001	0.000
	Dec	-12.81693	-2.80659	0.04882	0.00119	-0.00002	0.00000				
2027-02-25	GHA	483.93836	174.12746	-0.01710	-0.00046	0.00003	0.00000	56.780	-0.431	0.005	0.000
	Dec	-17.99730	-2.36315	0.06159	0.00094	-0.00001	0.00000				
2027-02-26	GHA	472.04565	173.97799	-0.01929	0.00010	0.00004	0.00000	55.967	-0.380	0.008	0.000
	Dec	-22.19799	-1.82932	0.07144	0.00070	-0.00002	0.00000				
2027-02-27	GHA	459.85824	173.83855	-0.01457	0.00067	0.00003	0.00000	55.277	-0.309	0.010	0.000
	Dec	-25.26162	-1.22863	0.07825	0.00043	-0.00002	0.00000				
2027-02-28	GHA	447.44986	173.76204	-0.00385	0.00107	0.00002	0.00000	54.740	-0.228	0.011	0.000
	Dec	-27.08038	-0.58744	0.08148	0.00010	-0.00002	0.00000				
2027-03-01	GHA	434.98593	173.78547	0.00984	0.00115	-0.00001	0.00000	54.370	-0.142	0.011	0.000
	Dec	-27.60362	0.06352	0.08066	-0.00024	-0.00002	0.00000				
2027-03-02	GHA	422.67678	173.91575	0.02225	0.00087	-0.00003	0.00000	54.170	-0.058	0.010	0.000
	Dec	-26.84421	0.69194	0.07592	-0.00054	0.00002	0.00000				
2027-03-03	GHA	410.71390	174.12774	0.02989	0.00038	-0.00003	0.00000	54.131	0.019	0.009	0.000
	Dec	-24.87628	1.26943	0.06808	-0.00075	-0.00001	0.00000				
2027-03-04	GHA	399.21675	174.37632	0.03135	-0.00013	-0.00003	0.00000	54.238	0.087	0.008	0.000

Moon 0h UT1		c0	c1	c2	c3	c4	c5	HP c0	HP c1	HP c2	HP c3
	Dec	-21.82299	1.77568	0.05827	-0.00087	-0.00001	0.00000				
2027-03-05	GHA	388.20972	174.61310	0.02711	-0.00056	-0.00002	0.00000	54.470	0.143	0.006	0.000
	Dec	-17.83944	2.19881	0.04741	-0.00093	0.00000	0.00000				
2027-03-06	GHA	377.62772	174.79775	0.01853	-0.00086	-0.00002	0.00000	54.800	0.186	0.005	0.000
	Dec	-13.09851	2.53258	0.03596	-0.00098	0.00000	0.00000				
2027-03-07	GHA	367.33626	174.90115	0.00697	-0.00106	-0.00001	0.00000	55.204	0.216	0.003	0.000
	Dec	-7.78343	2.77247	0.02391	-0.00104	0.00000	0.00000				
2027-03-08	GHA	357.15229	174.90355	-0.00661	-0.00120	-0.00001	0.00000	55.655	0.234	0.002	0.000
	Dec	-2.08756	2.91261	0.01095	-0.00113	-0.00001	0.00000				
2027-03-09	GHA	346.85997	174.79173	-0.02148	-0.00127	0.00000	0.00000	56.133	0.243	0.001	0.000
	Dec	3.78073	2.94370	-0.00343	-0.00127	-0.00001	0.00000				
2027-03-10	GHA	336.22302	174.55857	-0.03677	-0.00125	0.00001	0.00000	56.621	0.245	0.000	0.000
	Dec	9.59020	2.85214	-0.01979	-0.00146	-0.00001	0.00000				
2027-03-11	GHA	325.00033	174.20709	-0.05071	-0.00103	0.00002	0.00000	57.108	0.242	0.000	0.000
	Dec	15.07819	2.62020	-0.03856	-0.00167	-0.00001	0.00000				
2027-03-12	GHA	312.97559	173.76044	-0.05993	-0.00044	0.00005	0.00000	57.589	0.238	-0.001	0.000
	Dec	19.94466	2.22875	-0.05957	-0.00182	0.00000	0.00000				
2027-03-13	GHA	300.01145	173.27582	-0.05940	0.00060	0.00008	0.00000	58.058	0.231	-0.001	0.000
	Dec	23.85635	1.66477	-0.08133	-0.00176	0.00001	0.00000				
2027-03-14	GHA	286.12582	172.85089	-0.04459	0.00187	0.00007	0.00000	58.511	0.221	-0.002	0.000
	Dec	26.47214	0.93516	-0.10031	-0.00133	0.00004	0.00000				
2027-03-15	GHA	271.55422	172.60113	-0.01634	0.00272	0.00002	0.00000	58.935	0.202	-0.003	0.000
	Dec	27.49872	0.08212	-0.11144	-0.00047	0.00006	0.00000				
2027-03-16	GHA	256.73007	172.60199	0.01624	0.00254	-0.00005	0.00000	59.310	0.171	-0.005	0.000
	Dec	26.76628	-0.81397	-0.11072	0.00059	0.00006	0.00000				
2027-03-17	GHA	242.14916	172.83757	0.04074	0.00145	-0.00008	0.00000	59.606	0.123	-0.007	0.000
	Dec	24.28654	-1.65514	-0.09801	0.00148	0.00004	0.00000				
2027-03-18	GHA	228.18974	173.21101	0.05036	0.00017	-0.00007	0.00000	59.786	0.056	-0.010	0.000
	Dec	20.25567	-2.35805	-0.07683	0.00199	0.00002	0.00000				
2027-03-19	GHA	215.00879	173.60470	0.04643	-0.00076	-0.00004	0.00000	59.816	-0.028	-0.011	0.000
	Dec	15.00333	-2.87333	-0.05171	0.00215	0.00000	0.00000				
2027-03-20	GHA	202.55415	173.93041	0.03418	-0.00122	-0.00002	0.00000	59.668	-0.121	-0.012	0.000
	Dec	8.92468	-3.18389	-0.02602	0.00210	-0.00001	0.00000				
2027-03-21	GHA	190.63987	174.14228	0.01859	-0.00134	0.00000	0.00000	59.332	-0.214	-0.011	0.000
	Dec	2.42700	-3.29347	-0.00165	0.00195	-0.00001	0.00000				
2027-03-22	GHA	179.02314	174.22802	0.00305	-0.00122	0.00001	0.00000	58.822	-0.294	-0.009	0.000
	Dec	-4.10157	-3.21664	0.02049	0.00174	-0.00001	0.00000				
2027-03-23	GHA	527.45997	174.19784	-0.01010	-0.00094	0.00002	0.00000	58.174	-0.351	-0.005	0.000
	Dec	-10.30757	-2.97312	0.03998	0.00151	-0.00002	0.00000				
2027-03-24	GHA	515.74399	174.07885	-0.01887	-0.00050	0.00003	0.00000	57.440	-0.379	-0.002	0.000
	Dec	-15.87978	-2.58530	0.05653	0.00125	-0.00002	0.00000				
2027-03-25	GHA	503.73859	173.91361	-0.02142	0.00009	0.00004	0.00000	56.683	-0.376	0.002	0.000
	Dec	-20.55421	-2.07810	0.06975	0.00095	-0.00002	0.00000				
2027-03-26	GHA	491.40533	173.75716	-0.01661	0.00071	0.00004	0.00000	55.960	-0.344	0.006	0.000
	Dec	-24.12049	-1.48045	0.07903	0.00059	-0.00002	0.00000				
2027-03-27	GHA	478.81982	173.66690	-0.00516	0.00116	0.00002	0.00000	55.325	-0.290	0.008	0.000
	Dec	-26.43165	-0.82687	0.08364	0.00017	-0.00003	0.00000				
2027-03-28	GHA	466.15876	173.68449	0.00973	0.00126	-0.00001	0.00000	54.816	-0.218	0.010	0.000
	Dec	-27.41471	-0.15653	0.08323	-0.00024	-0.00002	0.00000				
2027-03-29	GHA	453.65076	173.81889	0.02335	0.00096	-0.00003	0.00000	54.461	-0.136	0.011	0.000
	Dec	-27.07524	0.49188	0.07830	-0.00057	-0.00002	0.00000				
2027-03-30	GHA	441.50587	174.04328	0.03181	0.00043	-0.00004	0.00000	54.276	-0.049	0.011	0.000
	Dec	-25.48939	1.08719	0.07018	-0.00077	-0.00001	0.00000				

Moon 0h UT1		c0	c1	c2	c3	c4	c5	HP c0	HP c1	HP c2	HP c3
2027-03-31	GHA	429.85627	174.30854	0.03353	-0.00013	-0.00003	0.00000	54.264	0.037	0.011	0.000
	Dec	-22.78396	1.61009	0.06041	-0.00085	0.00000	0.00000				
2027-04-01	GHA	418.73082	174.56233	0.02913	-0.00058	-0.00002	0.00000	54.418	0.117	0.010	0.000
	Dec	-19.11297	2.05232	0.05012	-0.00086	0.00000	0.00000				
2027-04-02	GHA	408.06241	174.76180	0.02019	-0.00089	-0.00002	0.00000	54.724	0.187	0.008	0.000
	Dec	-14.64020	2.41180	0.03973	-0.00087	0.00000	0.00000				
2027-04-03	GHA	397.71105	174.87685	0.00822	-0.00109	-0.00001	0.00000	55.157	0.243	0.006	0.000
	Dec	-9.53243	2.68701	0.02898	-0.00093	-0.00001	0.00000				
2027-04-04	GHA	387.48717	174.88763	-0.00576	-0.00123	-0.00001	0.00000	55.683	0.281	0.004	0.000
	Dec	-3.96310	2.87249	0.01717	-0.00105	-0.00001	0.00000				
2027-04-05	GHA	377.16866	174.78114	-0.02100	-0.00130	0.00000	0.00000	56.266	0.299	0.001	0.000
	Dec	1.87713	2.95630	0.00343	-0.00125	-0.00001	0.00000				
2027-04-06	GHA	366.51327	174.55042	-0.03665	-0.00128	0.00001	0.00000	56.864	0.297	-0.002	0.000
	Dec	7.76661	2.91940	-0.01311	-0.00151	-0.00002	0.00000				
2027-04-07	GHA	355.27415	174.19852	-0.05094	-0.00105	0.00002	0.00000	57.440	0.276	-0.004	0.000
	Dec	13.43970	2.73717	-0.03293	-0.00179	-0.00002	0.00000				
2027-04-08	GHA	343.22953	173.74891	-0.06042	-0.00045	0.00005	0.00000	57.958	0.241	-0.005	0.000
	Dec	18.58000	2.38417	-0.05565	-0.00197	-0.00001	0.00000				
2027-04-09	GHA	330.23829	173.26027	-0.05986	0.00062	0.00008	0.00000	58.397	0.197	-0.006	0.000
	Dec	22.82794	1.84406	-0.07926	-0.00190	0.00002	0.00000				
2027-04-10	GHA	316.31930	172.83354	-0.04450	0.00194	0.00008	0.00000	58.744	0.150	-0.006	0.000
	Dec	25.81424	1.12526	-0.09957	-0.00140	0.00005	0.00000				
2027-04-11	GHA	301.71699	172.58868	-0.01516	0.00282	0.00002	0.00000	58.997	0.103	-0.006	0.000
	Dec	27.22508	0.27593	-0.11111	-0.00047	0.00007	0.00000				
2027-04-12	GHA	286.88118	172.60374	0.01857	0.00262	-0.00005	0.00000	59.161	0.060	-0.005	0.000
	Dec	26.88354	-0.61662	-0.11011	0.00062	0.00006	0.00000				
2027-04-13	GHA	272.32492	172.86104	0.04374	0.00148	-0.00009	0.00000	59.241	0.020	-0.005	0.000
	Dec	24.80430	-1.45185	-0.09723	0.00146	0.00004	0.00000				
2027-04-14	GHA	258.43689	173.25897	0.05341	0.00016	-0.00007	0.00000	59.243	-0.019	-0.005	0.000
	Dec	21.18439	-2.15106	-0.07687	0.00187	0.00001	0.00000				
2027-04-15	GHA	245.37520	173.67572	0.04912	-0.00079	-0.00004	0.00000	59.166	-0.059	-0.005	0.000
	Dec	16.33978	-2.67433	-0.05383	0.00193	0.00000	0.00000				
2027-04-16	GHA	233.08242	174.02079	0.03632	-0.00127	-0.00002	0.00000	59.004	-0.103	-0.006	0.000
	Dec	10.63315	-3.01331	-0.03106	0.00185	-0.00001	0.00000				
2027-04-17	GHA	221.36383	174.24700	0.02000	-0.00141	0.00000	0.00000	58.749	-0.151	-0.006	0.000
	Dec	4.42687	-3.17507	-0.00958	0.00173	-0.00001	0.00000				
2027-04-18	GHA	209.96492	174.34030	0.00346	-0.00132	0.00001	0.00000	58.399	-0.199	-0.006	0.000
	Dec	-1.93536	-3.17036	0.01058	0.00163	-0.00001	0.00000				
2027-04-19	GHA	198.62537	174.30798	-0.01112	-0.00108	0.00002	0.00000	57.955	-0.244	-0.005	0.000
	Dec	-8.13060	-3.00904	0.02958	0.00153	-0.00001	0.00000				
2027-04-20	GHA	187.11573	174.17347	-0.02179	-0.00067	0.00003	0.00000	57.432	-0.278	-0.004	0.000
	Dec	-13.85511	-2.70072	0.04728	0.00141	-0.00001	0.00000				
2027-04-21	GHA	175.26958	173.97652	-0.02641	-0.00008	0.00004	0.00000	56.854	-0.298	-0.001	0.000
	Dec	-18.82707	-2.25812	0.06301	0.00120	-0.00002	0.00000				
2027-04-22	GHA	523.01679	173.77342	-0.02313	0.00063	0.00004	0.00000	56.254	-0.299	0.001	0.000
	Dec	-22.79728	-1.70163	0.07553	0.00087	-0.00002	0.00000				
2027-04-23	GHA	510.41028	173.62958	-0.01176	0.00123	0.00003	0.00000	55.674	-0.280	0.004	0.000
	Dec	-25.56850	-1.06310	0.08330	0.00041	-0.00003	0.00000				
2027-04-24	GHA	497.62645	173.60053	0.00493	0.00148	0.00000	0.00000	55.152	-0.240	0.006	0.000
	Dec	-27.01873	-0.38557	0.08518	-0.00010	-0.00003	0.00000				
2027-04-25	GHA	484.92156	173.70858	0.02167	0.00124	-0.00003	0.00000	54.726	-0.184	0.008	0.000
	Dec	-27.11776	0.28326	0.08125	-0.00054	-0.00002	0.00000				
2027-04-26	GHA	472.55292	173.93249	0.03329	0.00066	-0.00004	0.00000	54.428	-0.113	0.010	0.000

Moon 0h UT1		c0	c1	c2	c3	c4	c5	HP c0	HP c1	HP c2	HP c3
	Dec	-25.92551	0.90208	0.07297	-0.00081	-0.00001	0.00000				
2027-04-27	GHA	460.70138	174.21927	0.03725	0.00000	-0.00004	0.00000	54.282	-0.033	0.010	0.000
	Dec	-23.57003	1.44466	0.06250	-0.00091	0.00000	0.00000				
2027-04-28	GHA	449.43116	174.50770	0.03391	-0.00054	-0.00003	0.00000	54.301	0.053	0.011	0.000
	Dec	-20.21522	1.90110	0.05165	-0.00089	0.00000	0.00000				
2027-04-29	GHA	438.69258	174.74636	0.02511	-0.00091	-0.00002	0.00000	54.492	0.138	0.011	0.000
	Dec	-16.03293	2.27258	0.04132	-0.00083	0.00000	0.00000				
2027-04-30	GHA	428.34859	174.89932	0.01272	-0.00114	-0.00001	0.00000	54.850	0.219	0.010	0.000
	Dec	-11.18858	2.56356	0.03146	-0.00082	0.00000	0.00000				
2027-05-01	GHA	418.20359	174.94343	-0.00197	-0.00130	-0.00001	0.00000	55.359	0.289	0.008	0.000
	Dec	-5.84135	2.77524	0.02134	-0.00089	-0.00001	0.00000				
2027-05-02	GHA	408.02393	174.86334	-0.01824	-0.00141	-0.00001	0.00000	55.991	0.341	0.005	0.000
	Dec	-0.15559	2.90088	0.00976	-0.00106	-0.00001	0.00000				
2027-05-03	GHA	397.55050	174.64889	-0.03544	-0.00144	0.00000	0.00000	56.706	0.371	0.002	0.000
	Dec	5.68060	2.92329	-0.00467	-0.00136	-0.00002	0.00000				
2027-05-04	GHA	386.51092	174.29757	-0.05215	-0.00130	0.00002	0.00000	57.452	0.372	-0.002	0.000
	Dec	11.43376	2.81443	-0.02321	-0.00174	-0.00002	0.00000				
2027-05-05	GHA	374.64404	173.82463	-0.06519	-0.00079	0.00005	0.00000	58.172	0.344	-0.005	0.000
	Dec	16.80634	2.53889	-0.04631	-0.00210	-0.00002	0.00000				
2027-05-06	GHA	361.75282	173.28108	-0.06880	0.00029	0.00009	0.00000	58.807	0.288	-0.009	0.000
	Dec	21.43145	2.06430	-0.07257	-0.00222	0.00001	0.00000				
2027-05-07	GHA	347.79354	172.77011	-0.05620	0.00185	0.00010	0.00000	59.306	0.210	-0.011	0.000
	Dec	24.89777	1.38113	-0.09756	-0.00184	0.00004	0.00000				
2027-05-08	GHA	332.97213	172.43415	-0.02548	0.00315	0.00005	0.00000	59.636	0.119	-0.012	0.000
	Dec	26.81988	0.52708	-0.11418	-0.00084	0.00008	0.00000				
2027-05-09	GHA	317.76161	172.38859	0.01425	0.00324	-0.00004	0.00000	59.784	0.029	-0.011	0.000
	Dec	26.94456	-0.40414	-0.11627	0.00050	0.00008	0.00000				
2027-05-10	GHA	302.76408	172.64102	0.04671	0.00203	-0.00010	0.00000	59.759	-0.052	-0.009	0.000
	Dec	25.23936	-1.29024	-0.10335	0.00158	0.00005	0.00000				
2027-05-11	GHA	288.47725	173.08427	0.06127	0.00042	-0.00009	0.00000	59.587	-0.118	-0.007	0.000
	Dec	21.89976	-2.03058	-0.08088	0.00207	0.00001	0.00000				
2027-05-12	GHA	275.13584	173.57204	0.05859	-0.00077	-0.00005	0.00000	59.302	-0.165	-0.005	0.000
	Dec	17.27138	-2.57675	-0.05567	0.00207	-0.00001	0.00000				
2027-05-13	GHA	262.71072	173.99171	0.04529	-0.00136	-0.00002	0.00000	58.938	-0.198	-0.003	0.000
	Dec	11.74895	-2.92597	-0.03202	0.00185	-0.00002	0.00000				
2027-05-14	GHA	251.00166	174.28445	0.02760	-0.00154	0.00000	0.00000	58.521	-0.218	-0.002	0.000
	Dec	5.70835	-3.09730	-0.01122	0.00162	-0.00001	0.00000				
2027-05-15	GHA	239.73321	174.43190	0.00937	-0.00147	0.00001	0.00000	58.071	-0.231	-0.001	0.000
	Dec	-0.51668	-3.11241	0.00716	0.00146	-0.00001	0.00000				
2027-05-16	GHA	228.61810	174.43917	-0.00719	-0.00127	0.00002	0.00000	57.600	-0.240	-0.001	0.000
	Dec	-6.63002	-2.98684	0.02408	0.00137	0.00000	0.00000				
2027-05-17	GHA	217.39469	174.32640	-0.02038	-0.00091	0.00003	0.00000	57.115	-0.245	-0.001	0.000
	Dec	-12.35953	-2.72922	0.04023	0.00132	0.00000	0.00000				
2027-05-18	GHA	205.85594	174.12831	-0.02820	-0.00037	0.00004	0.00000	56.622	-0.248	0.000	0.000
	Dec	-17.44678	-2.34521	0.05562	0.00123	-0.00001	0.00000				
2027-05-19	GHA	193.88129	173.89684	-0.02842	0.00035	0.00005	0.00000	56.129	-0.245	0.001	0.000
	Dec	-21.64747	-1.84404	0.06932	0.00103	-0.00002	0.00000				
2027-05-20	GHA	181.46988	173.69898	-0.01976	0.00108	0.00004	0.00000	55.649	-0.234	0.002	0.000
	Dec	-24.74588	-1.24579	0.07959	0.00066	-0.00003	0.00000				
2027-05-21	GHA	528.75758	173.60222	-0.00361	0.00155	0.00001	0.00000	55.200	-0.214	0.003	0.000
	Dec	-26.58157	-0.58580	0.08451	0.00015	-0.00003	0.00000				
2027-05-22	GHA	515.99309	173.64907	0.01526	0.00151	-0.00002	0.00000	54.805	-0.181	0.005	0.000
	Dec	-27.07796	0.08819	0.08306	-0.00038	-0.00003	0.00000				

Moon 0h UT1		c0	c1	c2	c3	c4	c5	HP c0	HP c1	HP c2	HP c3
2027-05-23	GHA	503.46587	173.83666	0.03075	0.00101	-0.00004	0.00000	54.487	-0.136	0.006	0.000
	Dec	-26.25673	0.72711	0.07597	-0.00077	-0.00002	0.00000				
2027-05-24	GHA	491.41522	174.11930	0.03867	0.00030	-0.00004	0.00000	54.272	-0.078	0.008	0.000
	Dec	-24.22698	1.29377	0.06536	-0.00096	-0.00001	0.00000				
2027-05-25	GHA	479.96645	174.43158	0.03827	-0.00034	-0.00004	0.00000	54.183	-0.009	0.009	0.000
	Dec	-21.15368	1.76972	0.05361	-0.00098	0.00000	0.00000				
2027-05-26	GHA	469.11664	174.71270	0.03120	-0.00081	-0.00002	0.00000	54.241	0.068	0.010	0.000
	Dec	-17.22163	2.15293	0.04234	-0.00090	0.00001	0.00000				
2027-05-27	GHA	458.75706	174.91806	0.01963	-0.00110	-0.00001	0.00000	54.458	0.150	0.010	0.000
	Dec	-12.61006	2.45012	0.03211	-0.00081	0.00000	0.00000				
2027-05-28	GHA	448.70583	175.01868	0.00519	-0.00129	-0.00001	0.00000	54.841	0.233	0.010	0.000
	Dec	-7.48339	2.66860	0.02256	-0.00079	0.00000	0.00000				
2027-05-29	GHA	438.73376	174.99557	-0.01122	-0.00144	-0.00001	0.00000	55.385	0.311	0.009	0.000
	Dec	-1.99646	2.81011	0.01268	-0.00087	-0.00001	0.00000				
2027-05-30	GHA	428.57915	174.83488	-0.02914	-0.00154	0.00000	0.00000	56.074	0.376	0.007	0.000
	Dec	3.68978	2.86643	0.00102	-0.00109	-0.00002	0.00000				
2027-05-31	GHA	417.95678	174.52719	-0.04779	-0.00154	0.00001	0.00000	56.875	0.422	0.004	0.000
	Dec	9.38541	2.81661	0.01410	-0.00145	-0.00003	0.00000				
2027-06-01	GHA	406.57240	174.07403	-0.06503	-0.00127	0.00003	0.00000	57.739	0.439	0.000	0.000
	Dec	14.84535	2.62659	-0.03420	-0.00191	-0.00003	0.00000				
2027-06-02	GHA	394.15997	173.50437	-0.07595	-0.00045	0.00007	0.00000	58.603	0.420	-0.005	0.000
	Dec	19.74733	2.25422	-0.05959	-0.00230	-0.00002	0.00000				
2027-06-03	GHA	380.56070	172.89902	-0.07268	0.00109	0.00011	0.00000	59.390	0.362	-0.010	0.000
	Dec	23.69013	1.66508	-0.08772	-0.00230	0.00002	0.00000				
2027-06-04	GHA	365.84123	172.40176	-0.04841	0.00292	0.00010	0.00000	60.024	0.268	-0.014	0.000
	Dec	26.23723	0.86184	-0.11183	-0.00159	0.00007	0.00000				
2027-06-05	GHA	350.38377	172.17593	-0.00641	0.00384	0.00001	-0.00001	60.440	0.147	-0.016	0.000
	Dec	27.02159	-0.08667	-0.12280	-0.00017	0.00010	0.00000				
2027-06-06	GHA	334.82583	172.30212	0.03658	0.00307	-0.00009	0.00000	60.603	0.015	-0.016	0.000
	Dec	25.87848	-1.05027	-0.11544	0.00134	0.00008	0.00000				
2027-06-07	GHA	319.81870	172.71267	0.06285	0.00125	-0.00012	0.00000	60.508	-0.109	-0.014	0.000
	Dec	22.91868	-1.89082	-0.09309	0.00226	0.00003	0.00000				
2027-06-08	GHA	305.77327	173.24575	0.06756	-0.00037	-0.00008	0.00000	60.185	-0.211	-0.011	0.000
	Dec	18.48222	-2.52159	-0.06430	0.00244	-0.00001	0.00000				
2027-06-09	GHA	292.77794	173.74996	0.05690	-0.00129	-0.00004	0.00000	59.689	-0.282	-0.007	0.000
	Dec	13.01503	-2.92229	-0.03635	0.00218	-0.00002	0.00000				
2027-06-10	GHA	280.67892	174.13613	0.03908	-0.00161	-0.00001	0.00000	59.081	-0.323	-0.003	0.000
	Dec	6.95779	-3.11510	-0.01252	0.00180	-0.00002	0.00000				
2027-06-11	GHA	269.20215	174.37085	0.01962	-0.00160	0.00001	0.00000	58.420	-0.336	0.000	0.000
	Dec	0.69151	-3.13491	0.00701	0.00148	-0.00002	0.00000				
2027-06-12	GHA	258.04181	174.45361	0.00139	-0.00142	0.00002	0.00000	57.753	-0.329	0.002	0.000
	Dec	-5.46875	-3.01166	0.02345	0.00128	-0.00001	0.00000				
2027-06-13	GHA	246.90967	174.40153	-0.01387	-0.00110	0.00002	0.00000	57.113	-0.310	0.003	0.000
	Dec	-11.25719	-2.76454	0.03815	0.00118	0.00000	0.00000				
2027-06-14	GHA	235.56498	174.24494	-0.02446	-0.00063	0.00004	0.00000	56.519	-0.284	0.003	0.000
	Dec	-16.43699	-2.40376	0.05193	0.00112	0.00000	0.00000				
2027-06-15	GHA	223.84247	174.02933	-0.02831	0.00002	0.00005	0.00000	55.981	-0.255	0.004	0.000
	Dec	-20.78780	-1.93639	0.06471	0.00100	-0.00001	0.00000				
2027-06-16	GHA	211.68413	173.81621	-0.02366	0.00076	0.00005	0.00000	55.500	-0.225	0.004	0.000
	Dec	-24.10749	-1.37442	0.07534	0.00075	-0.00002	0.00000				
2027-06-17	GHA	199.16425	173.67481	-0.01062	0.00137	0.00003	0.00000	55.079	-0.195	0.004	0.000
	Dec	-26.22972	-0.74227	0.08197	0.00033	-0.00003	0.00000				
2027-06-18	GHA	186.48477	173.66084	0.00747	0.00157	0.00000	0.00000	54.720	-0.163	0.004	0.000

Moon 0h UT1		c0	c1	c2	c3	c4	c5	HP c0	HP c1	HP c2	HP c3
	Dec	-27.05182	-0.07900	0.08295	-0.00017	-0.00003	0.00000				
2027-06-19	GHA	173.92334	173.79224	0.02483	0.00126	-0.00003	0.00000	54.428	-0.128	0.005	0.000
	Dec	-26.55868	0.56799	0.07799	-0.00063	-0.00002	0.00000				
2027-06-20	GHA	521.74704	174.04097	0.03622	0.00062	-0.00004	0.00000	54.213	-0.087	0.006	0.000
	Dec	-24.82699	1.15570	0.06843	-0.00093	-0.00001	0.00000				
2027-06-21	GHA	510.13378	174.34845	0.03945	-0.00007	-0.00004	0.00000	54.086	-0.039	0.007	0.000
	Dec	-22.00508	1.65629	0.05655	-0.00103	0.00000	0.00000				
2027-06-22	GHA	499.13683	174.65109	0.03527	-0.00060	-0.00003	0.00000	54.064	0.018	0.008	0.000
	Dec	-18.27893	2.05971	0.04438	-0.00099	0.00001	0.00000				
2027-06-23	GHA	488.69355	174.89767	0.02575	-0.00096	-0.00002	0.00000	54.163	0.082	0.009	0.000
	Dec	-13.84088	2.36893	0.03309	-0.00089	0.00001	0.00000				
2027-06-24	GHA	478.65543	175.05339	0.01278	-0.00119	-0.00001	0.00000	54.399	0.154	0.009	0.000
	Dec	-8.87118	2.59225	0.02288	-0.00082	0.00000	0.00000				
2027-06-25	GHA	468.81724	175.09570	-0.00249	-0.00135	-0.00001	0.00000	54.783	0.230	0.010	0.000
	Dec	-3.53439	2.73648	0.01319	-0.00081	0.00000	0.00000				
2027-06-26	GHA	458.93573	175.00859	-0.01952	-0.00148	-0.00001	0.00000	55.319	0.306	0.009	0.000
	Dec	2.01243	2.80182	0.00296	-0.00091	-0.00001	0.00000				
2027-06-27	GHA	448.73906	174.77940	-0.03793	-0.00157	0.00000	0.00000	56.003	0.377	0.008	0.000
	Dec	7.60259	2.77812	-0.00931	-0.00115	-0.00002	0.00000				
2027-06-28	GHA	437.93486	174.40082	-0.05661	-0.00150	0.00001	0.00000	56.816	0.434	0.006	0.000
	Dec	13.03638	2.64235	-0.02530	-0.00153	-0.00003	0.00000				
2027-06-29	GHA	426.23032	173.88134	-0.07250	-0.00106	0.00005	0.00000	57.721	0.468	0.002	0.000
	Dec	18.05503	2.35872	-0.04640	-0.00198	-0.00003	0.00000				
2027-06-30	GHA	413.38372	173.26628	-0.07934	0.00004	0.00009	0.00000	58.661	0.468	-0.002	0.000
	Dec	22.32120	1.88576	-0.07241	-0.00230	-0.00001	0.00000				
2027-07-01	GHA	399.30312	172.66203	-0.06859	0.00182	0.00012	0.00000	59.561	0.427	-0.008	0.000
	Dec	25.42618	1.19649	-0.09959	-0.00212	0.00004	0.00000				
2027-07-02	GHA	384.16983	172.23194	-0.03597	0.00350	0.00007	0.00000	60.332	0.340	-0.014	0.000
	Dec	26.95146	0.31186	-0.11982	-0.00112	0.00009	0.00000				
2027-07-03	GHA	368.48772	172.12404	0.00953	0.00380	-0.00004	-0.00001	60.887	0.212	-0.018	0.000
	Dec	26.59220	-0.67464	-0.12400	0.00047	0.00010	0.00000				
2027-07-04	GHA	352.94384	172.36440	0.04817	0.00245	-0.00012	0.00000	61.159	0.059	-0.020	0.000
	Dec	24.28748	-1.61771	-0.10923	0.00191	0.00007	0.00000				
2027-07-05	GHA	338.12795	172.83458	0.06603	0.00055	-0.00011	0.00000	61.119	-0.099	-0.019	0.000
	Dec	20.26174	-2.38464	-0.08126	0.00262	0.00002	0.00000				
2027-07-06	GHA	324.32753	173.36293	0.06364	-0.00083	-0.00006	0.00000	60.779	-0.238	-0.015	0.000
	Dec	14.94357	-2.90684	-0.04930	0.00262	-0.00002	0.00000				
2027-07-07	GHA	311.52135	173.81879	0.04921	-0.00147	-0.00002	0.00000	60.195	-0.342	-0.010	0.000
	Dec	8.83105	-3.18122	-0.01997	0.00224	-0.00003	0.00000				
2027-07-08	GHA	299.49377	174.13785	0.03031	-0.00162	0.00000	0.00000	59.445	-0.404	-0.005	0.000
	Dec	2.38852	-3.24119	0.00418	0.00179	-0.00003	0.00000				
2027-07-09	GHA	287.95143	174.30407	0.01145	-0.00150	0.00001	0.00000	58.610	-0.427	-0.001	0.000
	Dec	-3.99713	-3.12849	0.02336	0.00143	-0.00002	0.00000				
2027-07-10	GHA	276.59727	174.32807	-0.00497	-0.00122	0.00002	0.00000	57.764	-0.416	0.003	0.000
	Dec	-10.01634	-2.87777	0.03890	0.00118	-0.00001	0.00000				
2027-07-11	GHA	265.17185	174.23628	-0.01724	-0.00080	0.00003	0.00000	56.962	-0.383	0.005	0.000
	Dec	-15.41765	-2.51249	0.05216	0.00103	-0.00001	0.00000				
2027-07-12	GHA	253.48250	174.06910	-0.02354	-0.00023	0.00004	0.00000	56.240	-0.337	0.006	0.000
	Dec	-19.98758	-2.04773	0.06380	0.00090	-0.00001	0.00000				
2027-07-13	GHA	241.43188	173.88142	-0.02217	0.00046	0.00004	0.00000	55.618	-0.285	0.007	0.000
	Dec	-23.54045	-1.49702	0.07353	0.00070	-0.00002	0.00000				
2027-07-14	GHA	229.04292	173.73755	-0.01269	0.00109	0.00003	0.00000	55.103	-0.231	0.007	0.000
	Dec	-25.92307	-0.88008	0.08014	0.00038	-0.00002	0.00000				

Moon 0h UT1		c0	c1	c2	c3	c4	c5	HP c0	HP c1	HP c2	HP c3
2027-07-15	GHA	216.46304	173.69547	0.00274	0.00142	0.00001	0.00000	54.691	-0.180	0.006	0.000
	Dec	-27.03268	-0.22783	0.08217	-0.00005	-0.00003	0.00000				
2027-07-16	GHA	203.92950	173.78490	0.01938	0.00129	-0.00002	0.00000	54.379	-0.132	0.006	0.000
	Dec	-26.83830	0.41947	0.07888	-0.00049	0.00003	0.00000				
2027-07-17	GHA	191.69802	173.99421	0.03206	0.00079	-0.00004	0.00000	54.160	-0.087	0.006	0.000
	Dec	-25.39131	1.02083	0.07087	-0.00082	-0.00002	0.00000				
2027-07-18	GHA	179.96530	174.27779	0.03772	0.00016	-0.00004	0.00000	54.029	-0.044	0.005	0.000
	Dec	-22.81644	1.54480	0.05982	-0.00099	-0.00001	0.00000				
2027-07-19	GHA	528.82164	174.57725	0.03618	-0.00039	-0.00003	0.00000	53.986	0.001	0.006	0.000
	Dec	-19.28676	1.97468	0.04760	-0.00102	0.00000	0.00000				
2027-07-20	GHA	518.24551	174.84060	0.02898	-0.00078	-0.00002	0.00000	54.034	0.048	0.006	0.000
	Dec	-14.99497	2.30710	0.03560	-0.00097	0.00000	0.00000				
2027-07-21	GHA	508.12545	175.03016	0.01797	-0.00103	-0.00001	0.00000	54.181	0.099	0.007	0.000
	Dec	-10.13200	2.54654	0.02439	-0.00090	0.00000	0.00000				
2027-07-22	GHA	498.28801	175.12116	0.00448	-0.00120	-0.00001	0.00000	54.436	0.156	0.007	0.000
	Dec	-4.87725	2.69947	0.01391	-0.00086	0.00000	0.00000				
2027-07-23	GHA	488.51879	175.09690	-0.01078	-0.00134	-0.00001	0.00000	54.810	0.218	0.008	0.000
	Dec	0.60027	2.76941	0.00352	-0.00089	0.00000	0.00000				
2027-07-24	GHA	478.57433	174.94474	-0.02744	-0.00143	0.00000	0.00000	55.310	0.283	0.008	0.000
	Dec	6.13236	2.75327	-0.00778	-0.00102	-0.00001	0.00000				
2027-07-25	GHA	468.18923	174.65561	-0.04488	-0.00145	0.00000	0.00000	55.940	0.346	0.008	0.000
	Dec	11.53545	2.63856	-0.02132	-0.00126	-0.00002	0.00000				
2027-07-26	GHA	457.08790	174.22931	-0.06134	-0.00124	0.00002	0.00000	56.691	0.404	0.006	0.000
	Dec	16.59017	2.40190	-0.03845	-0.00161	-0.00002	0.00000				
2027-07-27	GHA	445.01494	173.68810	-0.07281	-0.00058	0.00006	0.00000	57.542	0.445	0.004	0.000
	Dec	21.02098	2.01104	-0.05988	-0.00195	-0.00002	0.00000				
2027-07-28	GHA	431.80033	173.09740	-0.07261	0.00071	0.00010	0.00000	58.452	0.462	0.000	0.000
	Dec	24.48742	1.43497	-0.08435	-0.00206	0.00001	0.00000				
2027-07-29	GHA	417.46094	172.57855	-0.05420	0.00236	0.00010	0.00000	59.361	0.443	-0.005	0.000
	Dec	26.60745	0.66603	-0.10711	-0.00162	0.00005	0.00000				
2027-07-30	GHA	402.29000	172.28042	-0.01844	0.00342	0.00003	0.00000	60.188	0.380	-0.011	0.000
	Dec	27.03292	-0.25162	-0.12031	-0.00048	0.00009	0.00000				
2027-07-31	GHA	386.83329	172.29657	0.02182	0.00305	-0.00007	0.00000	60.844	0.271	-0.016	0.000
	Dec	25.56630	-1.21252	-0.11733	0.00098	0.00009	0.00000				
2027-08-01	GHA	371.70062	172.59424	0.04993	0.00154	-0.00011	0.00000	61.243	0.125	-0.020	0.000
	Dec	22.25556	-2.08207	-0.09800	0.00215	0.00005	0.00000				
2027-08-02	GHA	357.32709	173.03939	0.05858	-0.00004	-0.00008	0.00000	61.330	-0.040	-0.021	0.000
	Dec	17.39728	-2.75152	-0.06845	0.00267	0.00001	0.00000				
2027-08-03	GHA	343.85906	173.48661	0.05148	-0.00104	-0.00004	0.00000	61.088	-0.200	-0.019	0.000
	Dec	11.44878	-3.17013	-0.03629	0.00262	-0.00002	0.00000				
2027-08-04	GHA	331.19837	173.83986	0.03614	-0.00144	-0.00001	0.00000	60.552	-0.333	-0.014	0.000
	Dec	4.91408	-3.33996	-0.00679	0.00227	-0.00003	0.00000				
2027-08-05	GHA	319.11144	174.05848	0.01850	-0.00145	0.00001	0.00000	59.792	-0.423	-0.008	0.000
	Dec	-1.73923	-3.29283	0.01781	0.00183	-0.00003	0.00000				
2027-08-06	GHA	307.32314	174.13958	0.00215	-0.00124	0.00002	0.00000	58.897	-0.467	-0.003	0.000
	Dec	-8.11753	-3.06912	0.03744	0.00145	-0.00002	0.00000				
2027-08-07	GHA	295.57649	174.10312	-0.01061	-0.00086	0.00003	0.00000	57.958	-0.468	0.002	0.000
	Dec	-13.90494	-2.70535	0.05298	0.00115	-0.00002	0.00000				
2027-08-08	GHA	283.67110	173.98540	-0.01790	-0.00033	0.00004	0.00000	57.049	-0.437	0.006	0.000
	Dec	-18.85105	-2.23049	0.06531	0.00091	-0.00001	0.00000				
2027-08-09	GHA	271.49353	173.83684	-0.01813	0.00030	0.00004	0.00000	56.226	-0.384	0.008	0.000
	Dec	-22.75791	-1.66851	0.07474	0.00066	-0.00002	0.00000				
2027-08-10	GHA	259.04117	173.71691	-0.01081	0.00090	0.00003	0.00000	55.523	-0.319	0.009	0.000

Moon 0h UT1		c0	c1	c2	c3	c4	c5	HP c0	HP c1	HP c2	HP c3
	Dec	-25.47559	-1.04411	0.08081	0.00034	-0.00002	0.00000				
2027-08-11	GHA	246.42780	173.68074	0.00237	0.00124	0.00001	0.00000	54.955	-0.249	0.009	0.000
	Dec	-26.90881	-0.38756	0.08266	-0.00004	-0.00002	0.00000				
2027-08-12	GHA	233.85618	173.76021	0.01741	0.00120	0.00002	0.00000	54.527	-0.180	0.008	0.000
	Dec	-27.02892	0.26509	0.07982	-0.00043	-0.00002	0.00000				
2027-08-13	GHA	221.55727	173.95108	0.02961	0.00080	-0.00003	0.00000	54.233	-0.115	0.008	0.000
	Dec	-25.88046	0.87745	0.07272	-0.00074	-0.00002	0.00000				
2027-08-14	GHA	209.71994	174.21680	0.03585	0.00024	-0.00004	0.00000	54.062	-0.057	0.007	0.000
	Dec	-23.57440	1.42020	0.06264	-0.00092	-0.00001	0.00000				
2027-08-15	GHA	198.44286	174.50582	0.03550	-0.00028	-0.00003	0.00000	54.002	-0.005	0.006	0.000
	Dec	-20.26897	1.87558	0.05109	-0.00099	0.00000	0.00000				
2027-08-16	GHA	187.72290	174.76938	0.02972	-0.00066	-0.00002	0.00000	54.040	0.042	0.006	0.000
	Dec	-16.14672	2.23678	0.03923	-0.00098	0.00000	0.00000				
2027-08-17	GHA	177.47084	174.97058	0.02015	-0.00091	-0.00001	0.00000	54.166	0.085	0.005	0.000
	Dec	-11.39622	2.50407	0.02766	-0.00095	0.00000	0.00000				
2027-08-18	GHA	527.53627	175.08484	0.00813	-0.00108	-0.00001	0.00000	54.376	0.125	0.005	0.000
	Dec	-6.20248	2.68033	0.01645	-0.00093	0.00000	0.00000				
2027-08-19	GHA	517.72863	175.09615	-0.00550	-0.00119	-0.00001	0.00000	54.668	0.166	0.005	0.000
	Dec	-0.74552	2.76733	0.00527	-0.00094	0.00000	0.00000				
2027-08-20	GHA	507.83060	174.99349	-0.02030	-0.00127	0.00000	0.00000	55.043	0.209	0.005	0.000
	Dec	4.79461	2.76302	-0.00649	-0.00103	-0.00001	0.00000				
2027-08-21	GHA	497.60657	174.76964	-0.03569	-0.00128	0.00000	0.00000	55.505	0.254	0.006	0.000
	Dec	10.22811	2.65951	-0.01966	-0.00118	-0.00001	0.00000				
2027-08-22	GHA	486.81267	174.42416	-0.05043	-0.00114	0.00002	0.00000	56.058	0.299	0.006	0.000
	Dec	15.34239	2.44186	-0.03515	-0.00141	-0.00002	0.00000				
2027-08-23	GHA	475.21847	173.97215	-0.06178	-0.00069	0.00004	0.00000	56.701	0.343	0.005	0.000
	Dec	19.88849	2.08893	-0.05351	-0.00165	-0.00001	0.00000				
2027-08-24	GHA	462.65184	173.45843	-0.06507	0.00022	0.00007	0.00000	57.423	0.378	0.004	0.000
	Dec	23.57353	1.57882	-0.07423	-0.00177	0.00000	0.00000				
2027-08-25	GHA	449.07103	172.96912	-0.05503	0.00149	0.00008	0.00000	58.201	0.398	0.001	0.000
	Dec	26.07136	0.90199	-0.09461	-0.00156	0.00003	0.00000				
2027-08-26	GHA	434.64017	172.62084	-0.03012	0.00257	0.00005	0.00000	58.996	0.394	-0.002	0.000
	Dec	27.06635	0.08053	-0.10952	-0.00085	0.00006	0.00000				
2027-08-27	GHA	419.74415	172.51077	0.00291	0.00275	-0.00003	0.00000	59.749	0.356	-0.007	0.000
	Dec	26.33147	-0.81860	-0.11333	0.00025	0.00007	0.00000				
2027-08-28	GHA	404.88520	172.65405	0.03133	0.00186	-0.00008	0.00000	60.388	0.279	-0.012	0.000
	Dec	23.81100	-1.69364	-0.10347	0.00136	0.00006	0.00000				
2027-08-29	GHA	390.49873	172.97142	0.04566	0.00053	-0.00008	0.00000	60.835	0.165	-0.016	0.000
	Dec	19.65827	-2.44120	-0.08206	0.00214	0.00003	0.00000				
2027-08-30	GHA	376.81253	173.34182	0.04504	-0.00056	-0.00005	0.00000	61.024	0.022	-0.019	0.000
	Dec	14.20579	-2.98787	-0.05402	0.00247	0.00001	0.00000				
2027-08-31	GHA	363.82675	173.66344	0.03434	-0.00115	-0.00002	0.00000	60.917	-0.130	-0.019	0.000
	Dec	7.89235	-3.30072	-0.02423	0.00245	-0.00001	0.00000				
2027-09-01	GHA	351.38162	173.87878	0.01921	-0.00132	0.00000	0.00000	60.516	-0.269	-0.016	0.000
	Dec	1.18782	-3.38035	0.00390	0.00221	-0.00002	0.00000				
2027-09-02	GHA	339.24350	173.97032	0.00389	-0.00120	0.00001	0.00000	59.864	-0.378	-0.011	0.000
	Dec	-5.46140	-3.24837	0.02850	0.00188	-0.00002	0.00000				
2027-09-03	GHA	327.17323	173.94894	-0.00865	-0.00086	0.00003	0.00000	59.036	-0.445	-0.006	0.000
	Dec	-11.66316	-2.93643	0.04885	0.00151	-0.00002	0.00000				
2027-09-04	GHA	314.97486	173.84650	-0.01606	-0.00035	0.00004	0.00000	58.119	-0.468	0.000	0.000
	Dec	-17.09198	-2.47904	0.06487	0.00115	-0.00002	0.00000				
2027-09-05	GHA	302.53353	173.71179	-0.01651	0.00028	0.00004	0.00000	57.195	-0.452	0.004	0.000
	Dec	-21.49167	-1.91095	0.07651	0.00078	-0.00002	0.00000				

Moon 0h UT1		c0	c1	c2	c3	c4	c5	HP c0	HP c1	HP c2	HP c3
2027-09-06	GHA	289.84333	173.60395	-0.00940	0.00088	0.00003	0.00000	56.335	-0.406	0.007	0.000
	Dec	-24.67661	-1.26818	0.08347	0.00037	-0.00003	0.00000				
2027-09-07	GHA	277.01465	173.57822	0.00358	0.00123	0.00001	0.00000	55.587	-0.340	0.009	0.000
	Dec	-26.53611	-0.58973	0.08538	-0.00006	-0.00003	0.00000				
2027-09-08	GHA	264.24707	173.66661	0.01843	0.00118	-0.00002	0.00000	54.982	-0.264	0.010	0.000
	Dec	-27.03956	0.08367	0.08227	-0.00045	-0.00002	0.00000				
2027-09-09	GHA	251.76858	173.86496	0.03046	0.00078	-0.00003	0.00000	54.535	-0.183	0.010	0.000
	Dec	-26.23499	0.71483	0.07501	-0.00074	-0.00001	0.00000				
2027-09-10	GHA	239.76544	174.13700	0.03658	0.00023	-0.00003	0.00000	54.247	-0.105	0.010	0.000
	Dec	-24.23561	1.27624	0.06507	-0.00090	-0.00001	0.00000				
2027-09-11	GHA	228.33445	174.43167	0.03620	-0.00028	-0.00003	0.00000	54.110	-0.032	0.009	0.000
	Dec	-21.19742	1.75268	0.05396	-0.00094	0.00000	0.00000				
2027-09-12	GHA	217.47189	174.70099	0.03048	-0.00065	-0.00002	0.00000	54.111	0.032	0.007	0.000
	Dec	-17.29621	2.13910	0.04266	-0.00093	0.00000	0.00000				
2027-09-13	GHA	207.08974	174.90904	0.02112	-0.00089	-0.00001	0.00000	54.231	0.087	0.006	0.000
	Dec	-12.71197	2.43589	0.03156	-0.00092	0.00000	0.00000				
2027-09-14	GHA	197.04093	175.03245	0.00948	-0.00104	-0.00001	0.00000	54.450	0.132	0.005	0.000
	Dec	-7.62260	2.64428	0.02052	-0.00093	0.00000	0.00000				
2027-09-15	GHA	187.14105	175.05680	-0.00355	-0.00113	0.00000	0.00000	54.750	0.167	0.004	0.000
	Dec	-2.20558	2.76325	0.00913	-0.00098	0.00000	0.00000				
2027-09-16	GHA	177.18261	174.97318	-0.01744	-0.00118	0.00000	0.00000	55.114	0.196	0.003	0.000
	Dec	3.35569	2.78778	-0.00317	-0.00108	-0.00001	0.00000				
2027-09-17	GHA	526.94455	174.77697	-0.03158	-0.00116	0.00000	0.00000	55.528	0.219	0.003	0.000
	Dec	8.86315	2.70818	-0.01699	-0.00123	-0.00001	0.00000				
2027-09-18	GHA	516.20293	174.47039	-0.04479	-0.00101	0.00002	0.00000	55.986	0.239	0.002	0.000
	Dec	14.09480	2.51029	-0.03280	-0.00141	-0.00001	0.00000				
2027-09-19	GHA	504.75119	174.06960	-0.05469	-0.00059	0.00004	0.00000	56.481	0.257	0.002	0.000
	Dec	18.79726	2.17721	-0.05075	-0.00157	-0.00001	0.00000				
2027-09-20	GHA	492.43864	173.61547	-0.05749	0.00018	0.00006	0.00000	57.012	0.274	0.002	0.000
	Dec	22.68492	1.69435	-0.07004	-0.00161	0.00000	0.00000				
2027-09-21	GHA	479.22875	173.18159	-0.04913	0.00123	0.00007	0.00000	57.573	0.287	0.001	0.000
	Dec	25.45380	1.05923	0.08835	-0.00139	0.00003	0.00000				
2027-09-22	GHA	465.25779	172.86471	-0.02848	0.00214	0.00004	0.00000	58.155	0.294	0.000	0.000
	Dec	26.81877	0.29451	-0.10180	-0.00080	0.00005	0.00000				
2027-09-23	GHA	450.84587	172.74664	-0.00066	0.00235	-0.00002	0.00000	58.739	0.288	-0.002	0.000
	Dec	26.57295	-0.54412	-0.10634	0.00007	0.00006	0.00000				
2027-09-24	GHA	436.41754	172.84593	0.02433	0.00170	-0.00006	0.00000	59.293	0.264	-0.004	0.000
	Dec	24.64755	-1.37607	-0.10010	0.00095	0.00005	0.00000				
2027-09-25	GHA	422.35580	173.10393	0.03824	0.00060	-0.00007	0.00000	59.774	0.215	-0.008	0.000
	Dec	21.13964	-2.11875	-0.08441	0.00162	0.00003	0.00000				
2027-09-26	GHA	408.87959	173.42062	0.03920	-0.00039	-0.00005	0.00000	60.130	0.138	-0.011	0.000
	Dec	16.29365	-2.70894	-0.06248	0.00200	0.00002	0.00000				
2027-09-27	GHA	396.01105	173.70351	0.03045	-0.00101	-0.00003	0.00000	60.308	0.038	-0.014	0.000
	Dec	10.45420	-3.10961	-0.03742	0.00215	0.00000	0.00000				
2027-09-28	GHA	383.61954	173.89341	0.01660	-0.00125	-0.00001	0.00000	60.267	-0.079	-0.015	0.000
	Dec	4.01774	-3.30521	-0.01147	0.00215	0.00000	0.00000				
2027-09-29	GHA	371.49143	173.96595	0.00161	-0.00121	0.00001	0.00000	59.992	-0.196	-0.014	0.000
	Dec	-2.60360	-3.29503	0.01383	0.00205	-0.00001	0.00000				
2027-09-30	GHA	359.39294	173.92467	-0.01144	-0.00093	0.00002	0.00000	59.494	-0.299	-0.011	0.000
	Dec	-9.00740	-3.08923	0.03726	0.00184	-0.00002	0.00000				
2027-10-01	GHA	347.12088	173.79629	-0.01978	-0.00043	0.00004	0.00000	58.817	-0.375	-0.007	0.000
	Dec	-14.82102	-2.70737	0.05768	0.00154	-0.00002	0.00000				
2027-10-02	GHA	334.54675	173.62859	-0.02095	0.00025	0.00005	0.00000	58.024	-0.415	-0.003	0.000

Moon 0h UT1		c0	c1	c2	c3	c4	c5	HP c0	HP c1	HP c2	HP c3
	Dec	-19.72028	-2.17839	0.07386	0.00114	-0.00003	0.00000				
2027-10-03	GHA	321.65452	173.48528	-0.01365	0.00095	0.00004	0.00000	57.187	-0.418	0.002	0.000
	Dec	-23.44878	-1.54118	0.08456	0.00063	-0.00003	0.00000				
2027-10-04	GHA	308.55845	173.43074	0.00081	0.00140	0.00002	0.00000	56.376	-0.390	0.005	0.000
	Dec	-25.83740	-0.84391	0.08878	0.00007	-0.00003	0.00000				
2027-10-05	GHA	295.48090	173.50611	0.01802	0.00139	-0.00002	0.00000	55.648	-0.336	0.008	0.000
	Dec	-26.81866	-0.13929	0.08648	-0.00044	-0.00003	0.00000				
2027-10-06	GHA	282.68581	173.71079	0.03237	0.00095	-0.00004	0.00000	55.047	-0.264	0.010	0.000
	Dec	-26.42694	0.52467	0.07889	-0.00080	-0.00002	0.00000				
2027-10-07	GHA	270.39485	174.00459	0.03996	0.00031	-0.00004	0.00000	54.600	-0.182	0.011	0.000
	Dec	-24.77927	1.11406	0.06818	-0.00096	0.00000	0.00000				
2027-10-08	GHA	258.72799	174.32858	0.04001	-0.00028	-0.00003	0.00000	54.321	-0.097	0.011	0.000
	Dec	-22.04273	1.61270	0.05646	-0.00097	0.00000	0.00000				
2027-10-09	GHA	247.68909	174.62748	0.03398	-0.00070	-0.00002	0.00000	54.212	-0.013	0.010	0.000
	Dec	-18.40206	2.01849	0.04509	-0.00092	0.00000	0.00000				
2027-10-10	GHA	237.18597	174.86104	0.02396	-0.00095	-0.00001	0.00000	54.265	0.065	0.009	0.000
	Dec	-14.03843	2.33627	0.03445	-0.00086	0.00000	0.00000				
2027-10-11	GHA	227.06162	175.00430	0.01160	-0.00110	-0.00001	0.00000	54.463	0.133	0.008	0.000
	Dec	-9.12265	2.57101	0.02425	-0.00085	0.00000	0.00000				
2027-10-12	GHA	217.12031	175.04298	-0.00208	-0.00118	0.00000	0.00000	54.785	0.187	0.006	0.000
	Dec	-3.81948	2.72342	0.01375	-0.00091	-0.00001	0.00000				
2027-10-13	GHA	207.14421	174.96890	-0.01651	-0.00122	0.00000	0.00000	55.201	0.227	0.004	0.000
	Dec	1.70131	2.78765	0.00207	-0.00105	-0.00001	0.00000				
2027-10-14	GHA	196.90363	174.77836	-0.03108	-0.00119	0.00000	0.00000	55.681	0.251	0.002	0.000
	Dec	7.25122	2.75085	-0.01163	-0.00125	-0.00001	0.00000				
2027-10-15	GHA	186.16776	174.47448	-0.04460	-0.00103	0.00002	0.00000	56.194	0.261	0.000	0.000
	Dec	12.60978	2.59409	-0.02797	-0.00148	-0.00001	0.00000				
2027-10-16	GHA	174.72514	174.07450	-0.05464	-0.00059	0.00004	0.00000	56.713	0.257	-0.001	0.000
	Dec	17.51543	2.29571	-0.04697	-0.00167	-0.00001	0.00000				
2027-10-17	GHA	522.42302	173.62113	-0.05730	0.00021	0.00006	0.00000	57.214	0.244	-0.002	0.000
	Dec	21.66635	1.83808	-0.06750	-0.00171	0.00001	0.00000				
2027-10-18	GHA	509.22745	173.19072	-0.04839	0.00129	0.00007	0.00000	57.682	0.224	-0.003	0.000
	Dec	24.73967	1.21916	-0.08675	-0.00144	0.00003	0.00000				
2027-10-19	GHA	495.28335	172.88318	-0.02685	0.00222	0.00004	0.00000	58.109	0.202	-0.003	0.000
	Dec	26.43608	0.46581	-0.10044	-0.00079	0.00005	0.00000				
2027-10-20	GHA	480.92441	172.78190	0.00187	0.00242	-0.00002	0.00000	58.490	0.179	-0.003	0.000
	Dec	26.54474	-0.36061	-0.10458	0.00011	0.00006	0.00000				
2027-10-21	GHA	466.58880	172.90383	0.02738	0.00172	-0.00006	0.00000	58.823	0.154	-0.003	0.000
	Dec	25.00190	-1.17669	-0.09796	0.00096	0.00005	0.00000				
2027-10-22	GHA	452.66762	173.18661	0.04131	0.00058	-0.00007	0.00000	59.102	0.125	-0.004	0.000
	Dec	21.90893	-1.90362	-0.08280	0.00151	0.00002	0.00000				
2027-10-23	GHA	439.38058	173.52687	0.04198	-0.00042	-0.00005	0.00000	59.316	0.088	-0.005	0.000
	Dec	17.50060	-2.48807	-0.06287	0.00177	0.00001	0.00000				
2027-10-24	GHA	426.74538	173.83020	0.03275	-0.00105	-0.00003	0.00000	59.448	0.041	-0.007	0.000
	Dec	12.08976	-2.90465	-0.04116	0.00184	0.00000	0.00000				
2027-10-25	GHA	414.62364	174.03591	0.01820	-0.00132	-0.00001	0.00000	59.473	-0.018	-0.008	0.000
	Dec	6.02106	-3.14566	-0.01909	0.00184	0.00000	0.00000				
2027-10-26	GHA	402.79002	174.11698	0.00205	-0.00133	0.00001	0.00000	59.368	-0.087	-0.009	0.000
	Dec	-0.35322	-3.21027	0.00292	0.00183	0.00000	0.00000				
2027-10-27	GHA	390.99168	174.07222	-0.01287	-0.00112	0.00002	0.00000	59.119	-0.162	-0.009	0.000
	Dec	-6.68112	-3.09945	0.02473	0.00180	0.00000	0.00000				
2027-10-28	GHA	378.99507	173.92193	-0.02393	-0.00068	0.00004	0.00000	58.722	-0.234	-0.008	0.000
	Dec	-12.61480	-2.81666	0.04579	0.00170	-0.00001	0.00000				

Moon 0h UT1		c0	c1	c2	c3	c4	c5	HP c0	HP c1	HP c2	HP c3
2027-10-29	GHA	366.62914	173.70853	-0.02823	0.00000	0.00005	0.00000	58.192	-0.294	-0.006	0.000
	Dec	-17.81973	-2.37236	0.06486	0.00146	-0.00002	0.00000				
2027-10-30	GHA	353.83007	173.49649	-0.02333	0.00082	0.00005	0.00000	57.562	-0.334	-0.003	0.000
	Dec	-21.99482	-1.79014	0.07994	0.00103	-0.00003	0.00000				
2027-10-31	GHA	340.67672	173.36209	-0.00905	0.00151	0.00003	0.00000	56.877	-0.348	0.000	0.000
	Dec	-24.90341	-1.11108	0.08877	0.00042	-0.00004	0.00000				
2027-11-01	GHA	327.39004	173.36790	0.01088	0.00172	-0.00001	0.00000	56.191	-0.336	0.003	0.000
	Dec	-26.40723	-0.39178	0.08991	-0.00023	-0.00004	0.00000				
2027-11-02	GHA	314.27542	173.53333	0.02983	0.00135	-0.00004	0.00000	55.553	-0.299	0.006	0.000
	Dec	-26.48707	0.30684	0.08382	-0.00075	-0.00003	0.00000				
2027-11-03	GHA	301.62399	173.82532	0.04190	0.00063	-0.00005	0.00000	55.010	-0.242	0.008	0.000
	Dec	-25.23563	0.93541	0.07283	-0.00104	-0.00001	0.00000				
2027-11-04	GHA	289.62458	174.17778	0.04492	-0.00011	-0.00004	0.00000	54.598	-0.169	0.010	0.000
	Dec	-22.82276	1.46667	0.05990	0.00109	0.00000	0.00000				
2027-11-05	GHA	278.32825	174.52189	0.04015	-0.00065	-0.00003	0.00000	54.342	-0.086	0.011	0.000
	Dec	-19.45066	1.89493	0.04732	-0.00100	0.00001	0.00000				
2027-11-06	GHA	267.66386	174.80536	0.03014	-0.00098	-0.00002	0.00000	54.257	0.001	0.011	0.000
	Dec	-15.31866	2.22771	0.03609	-0.00088	0.00001	0.00000				
2027-11-07	GHA	257.47582	174.99576	0.01714	-0.00117	-0.00001	0.00000	54.347	0.088	0.011	0.000
	Dec	-10.60662	2.47604	0.02614	-0.00079	0.00000	0.00000				
2027-11-08	GHA	247.55881	175.07502	0.00250	-0.00127	0.00000	0.00000	54.605	0.169	0.010	0.000
	Dec	-5.47530	2.64742	0.01670	-0.00079	0.00000	0.00000				
2027-11-09	GHA	237.67984	175.03300	-0.01311	-0.00133	0.00000	0.00000	55.015	0.239	0.008	0.000
	Dec	-0.07792	2.74160	0.00666	-0.00090	-0.00001	0.00000				
2027-11-10	GHA	227.59003	174.86377	-0.02922	-0.00134	0.00000	0.00000	55.549	0.293	0.006	0.000
	Dec	5.42226	2.74864	-0.00527	-0.00111	-0.00002	0.00000				
2027-11-11	GHA	217.03351	174.56664	-0.04488	-0.00124	0.00001	0.00000	56.170	0.326	0.003	0.000
	Dec	10.83192	2.64852	-0.02028	-0.00141	-0.00002	0.00000				
2027-11-12	GHA	205.76400	174.15297	-0.05790	-0.00087	0.00003	0.00000	56.834	0.335	0.000	0.000
	Dec	15.90939	2.41328	-0.03908	-0.00172	-0.00002	0.00000				
2027-11-13	GHA	193.58167	173.65948	-0.06409	-0.00008	0.00006	0.00000	57.490	0.319	-0.003	0.000
	Dec	20.35486	2.01379	-0.06113	-0.00192	0.00000	0.00000				
2027-11-14	GHA	180.39866	173.16258	-0.05797	0.00115	0.00008	0.00000	58.092	0.281	-0.006	0.000
	Dec	23.82099	1.43351	-0.08369	-0.00177	0.00002	0.00000				
2027-11-15	GHA	526.31975	172.77659	-0.03631	0.00240	0.00006	0.00000	58.599	0.224	-0.008	0.000
	Dec	25.95752	0.68812	-0.10152	-0.00112	0.00006	0.00000				
2027-11-16	GHA	511.68285	172.61400	-0.00345	0.00290	-0.00001	0.00000	58.982	0.158	-0.009	0.000
	Dec	26.49108	-0.16067	-0.10881	-0.00006	0.00007	0.00000				
2027-11-17	GHA	496.98872	172.71857	0.02845	0.00225	-0.00007	0.00000	59.231	0.091	-0.008	0.000
	Dec	25.31042	-1.01504	-0.10291	0.00101	0.00006	0.00000				
2027-11-18	GHA	482.72402	173.03285	0.04778	0.00093	-0.00009	0.00000	59.350	0.029	-0.007	0.000
	Dec	22.50495	-1.77652	-0.08627	0.00169	0.00003	0.00000				
2027-11-19	GHA	469.19134	173.43724	0.05114	-0.00031	-0.00006	0.00000	59.354	-0.024	-0.006	0.000
	Dec	18.32988	-2.38015	-0.06425	0.00192	0.00000	0.00000				
2027-11-20	GHA	456.45247	173.81648	0.04231	-0.00108	-0.00003	0.00000	59.261	-0.067	-0.005	0.000
	Dec	13.12833	-2.80236	-0.04141	0.00186	-0.00001	0.00000				
2027-11-21	GHA	444.37756	174.09575	0.02691	-0.00143	-0.00001	0.00000	59.092	-0.102	-0.004	0.000
	Dec	7.26133	-3.04675	-0.01994	0.00172	-0.00001	0.00000				
2027-11-22	GHA	432.72893	174.24091	0.00929	-0.00148	0.00000	0.00000	58.857	-0.133	-0.004	0.000
	Dec	1.07206	-3.12596	-0.00006	0.00161	-0.00001	0.00000				
2027-11-23	GHA	421.23024	174.24622	-0.00770	-0.00133	0.00002	0.00000	58.562	-0.162	-0.004	0.000
	Dec	-5.12013	-3.05048	0.01884	0.00155	0.00000	0.00000				
2027-11-24	GHA	409.61413	174.12592	-0.02177	-0.00098	0.00003	0.00000	58.210	-0.191	-0.004	0.000

Moon 0h UT1		c0	c1	c2	c3	c4	c5	HP c0	HP c1	HP c2	HP c3
	Dec	-11.01159	-2.82555	0.03735	0.00153	0.00000	0.00000				
2027-11-25	GHA	397.66068	173.91350	-0.03032	-0.00040	0.00004	0.00000	57.799	-0.220	-0.003	0.000
	Dec	-16.30647	-2.45434	0.05532	0.00145	-0.00001	0.00000				
2027-11-26	GHA	385.23908	173.66478	-0.03044	0.00041	0.00006	0.00000	57.333	-0.246	-0.003	0.000
	Dec	-20.71966	-1.94531	0.07153	0.00122	-0.00002	0.00000				
2027-11-27	GHA	372.35109	173.45567	-0.02033	0.00126	0.00005	0.00000	56.822	-0.264	-0.002	0.000
	Dec	-23.99646	-1.32136	0.08367	0.00077	-0.00004	0.00000				
2027-11-28	GHA	359.15546	173.36442	-0.00153	0.00180	0.00002	0.00000	56.285	-0.272	0.000	0.000
	Dec	-25.94793	-0.62549	0.08919	0.00014	-0.00004	0.00000				
2027-11-29	GHA	345.94133	173.43962	0.02020	0.00172	-0.00002	0.00000	55.747	-0.264	0.002	0.000
	Dec	-26.48825	0.08322	0.08686	-0.00051	-0.00004	0.00000				
2027-11-30	GHA	333.04150	173.67505	0.03756	0.00111	-0.00005	0.00000	55.242	-0.240	0.004	0.000
	Dec	-25.65256	0.74485	0.07775	-0.00097	0.00002	0.00000				
2027-12-01	GHA	320.72433	174.01476	0.04593	0.00029	-0.00005	0.00000	54.803	-0.198	0.006	0.000
	Dec	-23.58076	1.31607	0.06473	-0.00116	0.00000	0.00000				
2027-12-02	GHA	309.12320	174.38330	0.04499	-0.00041	-0.00004	0.00000	54.463	-0.140	0.008	0.000
	Dec	-20.47493	1.77818	0.05085	-0.00113	0.00001	0.00000				
2027-12-03	GHA	298.22795	174.71485	0.03710	-0.00087	-0.00002	0.00000	54.253	-0.069	0.010	0.000
	Dec	-16.55311	2.13301	0.03810	-0.00099	0.00001	0.00000				
2027-12-04	GHA	287.91786	174.96493	0.02497	-0.00113	-0.00001	0.00000	54.194	0.012	0.011	0.000
	Dec	-12.01805	2.39291	0.02713	-0.00084	0.00001	0.00000				
2027-12-05	GHA	278.00267	175.10782	0.01050	-0.00127	-0.00001	0.00000	54.303	0.098	0.011	0.000
	Dec	-7.04603	2.57118	0.01759	-0.00076	0.00000	0.00000				
2027-12-06	GHA	268.25275	175.12904	-0.00536	-0.00137	-0.00001	0.00000	54.587	0.185	0.011	0.000
	Dec	-1.79155	2.67577	0.00854	-0.00077	0.00000	0.00000				
2027-12-07	GHA	258.41513	175.01931	-0.02219	-0.00143	0.00000	0.00000	55.041	0.268	0.010	0.000
	Dec	3.59794	2.70551	-0.00133	-0.00090	-0.00001	0.00000				
2027-12-08	GHA	248.22142	174.77254	-0.03951	-0.00144	0.00000	0.00000	55.650	0.339	0.008	0.000
	Dec	8.96154	2.64796	-0.01351	-0.00115	-0.00002	0.00000				
2027-12-09	GHA	237.39726	174.38959	-0.05592	-0.00126	0.00002	0.00000	56.382	0.391	0.005	0.000
	Dec	14.10153	2.47878	-0.02942	-0.00151	-0.00002	0.00000				
2027-12-10	GHA	225.68658	173.88952	-0.06812	-0.00070	0.00005	0.00000	57.192	0.416	0.001	0.000
	Dec	18.76175	2.16443	-0.04982	-0.00188	-0.00002	0.00000				
2027-12-11	GHA	212.90594	173.32796	-0.07030	0.00043	0.00009	0.00000	58.020	0.409	-0.003	0.000
	Dec	22.61776	1.67168	-0.07366	-0.00204	0.00000	0.00000				
2027-12-12	GHA	199.03359	172.81150	-0.05611	0.00196	0.00010	0.00000	58.798	0.365	-0.008	0.000
	Dec	25.29661	0.98789	-0.09671	-0.00170	0.00004	0.00000				
2027-12-13	GHA	184.29893	172.48016	-0.02463	0.00315	0.00004	0.00000	59.454	0.288	-0.012	0.000
	Dec	26.44407	0.14692	-0.11182	-0.00072	0.00008	0.00000				
2027-12-14	GHA	529.18541	172.43882	0.01420	0.00310	-0.00005	0.00000	59.929	0.185	-0.014	0.000
	Dec	25.83162	-0.76001	-0.11257	0.00061	0.00008	0.00000				
2027-12-15	GHA	514.28161	172.68274	0.04455	0.00183	-0.00010	0.00000	60.183	0.069	-0.015	0.000
	Dec	23.44869	-1.61092	-0.09822	0.00171	0.00005	0.00000				
2027-12-16	GHA	500.05399	173.10001	0.05702	0.00027	-0.00009	0.00000	60.209	-0.042	-0.013	0.000
	Dec	19.51388	-2.30372	-0.07409	0.00222	0.00001	0.00000				
2027-12-17	GHA	486.70534	173.54793	0.05300	-0.00085	-0.00005	0.00000	60.028	-0.137	-0.010	0.000
	Dec	14.39914	-2.78850	-0.04712	0.00222	-0.00001	0.00000				
2027-12-18	GHA	474.18481	173.91985	0.03903	-0.00140	-0.00002	0.00000	59.681	-0.207	-0.007	0.000
	Dec	8.52679	-3.06285	-0.02192	0.00196	-0.00002	0.00000				
2027-12-19	GHA	462.28090	174.16125	0.02107	-0.00155	0.00000	0.00000	59.219	-0.252	-0.004	0.000
	Dec	2.29692	-3.14883	-0.00007	0.00168	-0.00002	0.00000				
2027-12-20	GHA	450.71387	174.25656	0.00292	-0.00145	0.00001	0.00000	58.692	-0.273	-0.001	0.000
	Dec	-3.94017	-3.07257	0.01877	0.00147	-0.00001	0.00000				

Moon 0h UT1		c0	c1	c2	c3	c4	c5	HP c0	HP c1	HP c2	HP c3
2027-12-21	GHA	439.19804	174.21432	-0.01300	-0.00118	0.00002	0.00000	58.139	-0.278	0.000	0.000
	Dec	-9.88077	-2.85397	0.03567	0.00135	-0.00001	0.00000				
2027-12-22	GHA	427.48299	174.06104	-0.02449	-0.00070	0.00004	0.00000	57.588	-0.272	0.001	0.000
	Dec	-15.25299	-2.50514	0.05140	0.00127	-0.00001	0.00000				
2027-12-23	GHA	415.39003	173.84222	-0.02901	-0.00002	0.00005	0.00000	57.053	-0.262	0.001	0.000
	Dec	-19.80521	-2.03499	0.06590	0.00113	-0.00001	0.00000				
2027-12-24	GHA	402.85151	173.62317	-0.02431	0.00080	0.00005	0.00000	56.542	-0.249	0.002	0.000
	Dec	-23.30789	-1.45764	0.07793	0.00084	-0.00002	0.00000				
2027-12-25	GHA	389.94301	173.48006	-0.01025	0.00149	0.00003	0.00000	56.057	-0.236	0.002	0.000
	Dec	-25.57284	-0.80110	0.08536	0.00037	-0.00003	0.00000				
2027-12-26	GHA	376.88220	173.47571	0.00957	0.00172	0.00000	0.00000	55.600	-0.221	0.002	0.000
	Dec	-26.48511	-0.11041	0.08627	-0.00022	-0.00004	0.00000				
2027-12-27	GHA	363.97300	173.63105	0.02866	0.00138	-0.00004	0.00000	55.176	-0.202	0.003	0.000
	Dec	-26.03090	0.55964	0.08031	-0.00075	-0.00003	0.00000				
2027-12-28	GHA	351.50887	173.91524	0.04114	0.00067	-0.00005	0.00000	54.797	-0.177	0.004	0.000
	Dec	-24.30225	1.15967	0.06913	-0.00108	-0.00001	0.00000				
2027-12-29	GHA	339.68496	174.26382	0.04471	-0.00006	-0.00004	0.00000	54.476	-0.143	0.005	0.000
	Dec	-21.47254	1.65862	0.05545	-0.00117	0.00000	0.00000				
2027-12-30	GHA	328.56074	174.60841	0.04047	-0.00061	-0.00003	0.00000	54.233	-0.099	0.006	0.000
	Dec	-17.75565	2.04681	0.04172	-0.00110	0.00001	0.00000				
2027-12-31	GHA	318.07331	174.89615	0.03087	-0.00096	-0.00002	0.00000	54.090	-0.043	0.008	0.000
	Dec	-13.36850	2.33001	0.02932	-0.00096	0.00001	0.00000				